

US Semiconductors

Rise of the Agents: raising CPU TAM to \$170bn

Price Objective Change

CPU reborn: Agentic AI drives 5x demand

Based on our analysis and recent industry discussions at BofA Global Tech Conference (see our [conf takeaways](#)), we raise our CY2030E server CPU TAM to \$170bn+ (from \$125bn), implying nearly 5x growth and +37% CAGR over CY25-30E (vs. +29% prior). We view the emergence of agentic AI as a powerful demand accelerant that expands the CPU opportunity and lifts both x86 incumbents and ARM challengers. Against this backdrop, we raise our **AMD** PO to \$560 from \$500 on higher CPU/GPU estimates, **ARM** to \$335 from \$245 on greater I-t chiplet potential; **INTC** is double-upgraded to Buy with a \$135 PO (see our), with higher estimates now reflecting both n-t CPU and I-t foundry upsides. **NVDA** remains our top sector pick on full-stack AI leadership, benefiting from tight CPU-GPU-networking integration, also poised to benefit from agentic CPU. **QCOM** is expected to announce new AI CPUs at its Jun-24 AI Day in NYC, but we maintain Underperform given tough competition and limited SAM.

Agentic workloads shift complexity to CPUs from XPU

Agentic AI differs from traditional genAI by shifting from a single prompt-response workflow to a multi-step system that plans, reasons, retrieves info, uses tools, executes code simultaneously. While XPUs remain critical for inference, many orchestration and decision-making functions are latency-sensitive, sequential, and I/O-intensive—making them better suited for CPUs. Our new CPU model uniquely segments the \$170bn TAM by application (1. standalone agentic CPU; 2. head/compute node CPU for AI clusters; 3. traditional/laaS), as well as by vendor (AMD, INTC, NVDA, ARM merchant, ARM custom). We expect head/compute nodes in AI clusters to use higher frequency and stronger but fewer cores, while agentic and traditional applications will rely on higher core counts.

Double-upgrading INTC, while AMD top CPU pick

Our **INTC** double-upgrade is based on higher confidence in INTC's opportunity to help address industry constraints in leading edge wafers/packaging, plus larger agentic CPU TAM. We now see CY30 EPS power \$6+ vs \$3-4 prior, though execution is key. Recent CDNS 14A node IP sign-up and Terafab engagements are additional supportive data points. For **AMD**, we raise est/PO to \$560, top CPU pick on incumbency + pipeline + upcoming AI day (Venice launch). **NVDA** remains top sector pick on full-stack leadership.

ARM/QCOM: CPU share gainers but flag val'n/competition

For **ARM**, we now use a FY31/CY30E sum-of-parts valuation, see total company value of ~\$335 (\$229 IP + \$106 chip), and we see the stock as fairly valued. **QCOM** has potential to announce new AI CPUs and lead customers at its Jun-24 AI Day, but we flag tough CPU competition amid already established incumbents.

Key debate: TAM substitution vs expansion

Debate centers on whether CPUs take share from GPUs/accelerators or contribute to overall AI TAM expansion. We see expansion dominating: CPU demand scales with system complexity, as is the case with reasoning/agentic AI. Additional debates include durability of the 30%+ AI capex CAGR and mix shifts to custom silicon (CPUs and XPUs).

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Exhibit 1: PO/Rating Changes

PO/Rating Changes

PO/Rating Changes			
Ticker	OLD	NEW	RATING
AMD	\$500	\$560	BUY
ARM	\$245	\$335	NEUTRAL
INTC	\$96	\$135	BUY from U/P
NVDA	\$350	\$350	BUY
QCOM	\$165	\$165	U/P

Source: BofA Global Research

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For detailed company analysis, PO/rating change and valuation discussion, see pg. 10-17.

See glossary on page 23.

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AI Compute Table of Hierarchy

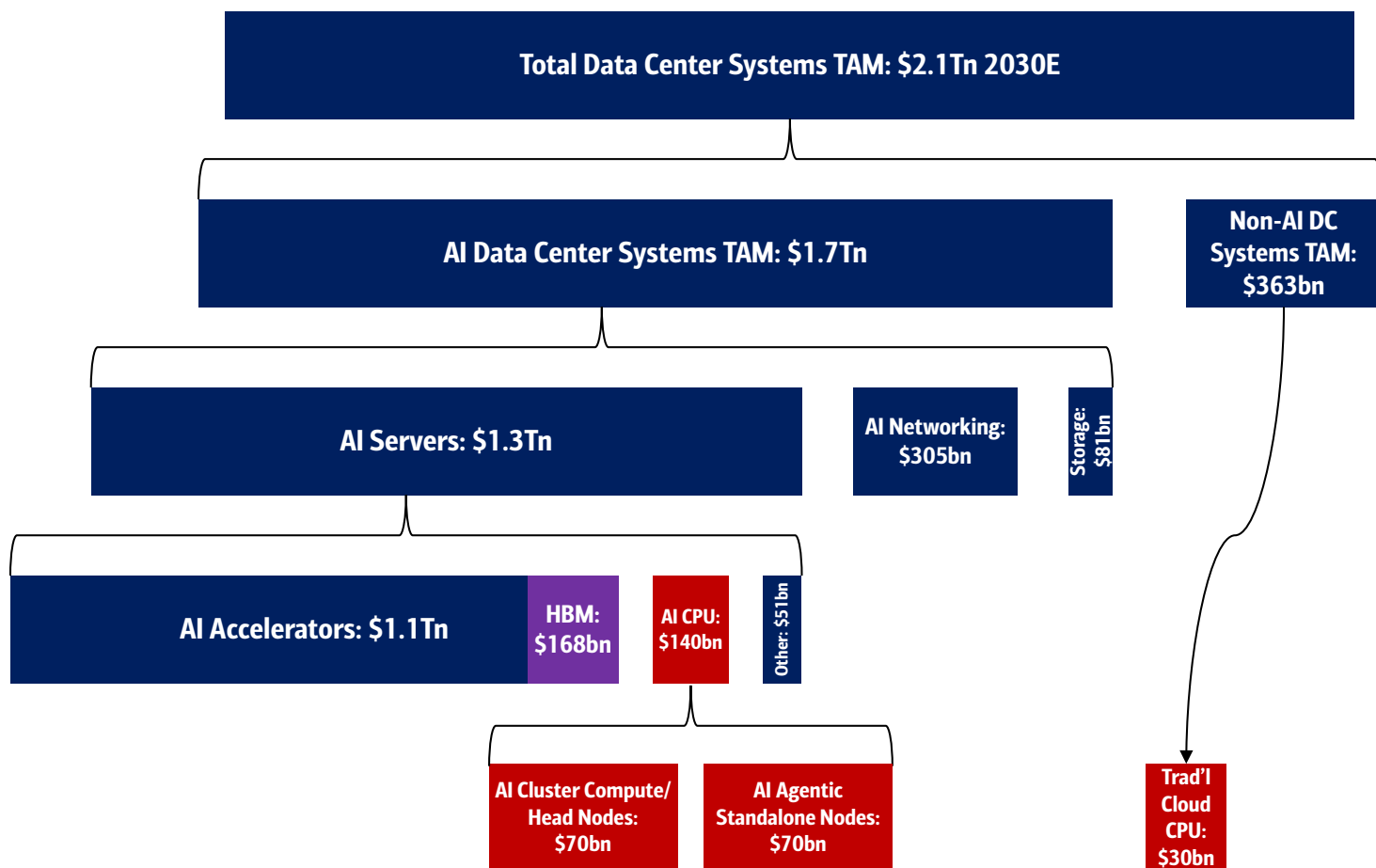
Below, we present our latest table of hierarchy for global data center systems TAM of ~\$2.1Tn, of which CPUs make up ~\$170bn of the total value.

Within this, we break out CPUs in three different categories (more detail in following pages):

1. **Traditional/on-prem/multi-tenant cloud CPUs:** ~\$30bn
2. **AI cluster compute/head node CPUs:** ~\$70bn
3. **AI agentic standalone node CPUs:** ~\$70bn

Exhibit 2: Of the total \$2.1Tn CY30 DC Systems TAM, we see AI CPUs representing ~\$140bn of the value (split roughly 50/50 across compute/head node and agentic AI node), non-AI CPUs ~\$30bn

Data Center Systems TAM – Breakout by: AI vs. non-AI; AI CPUs vs. non-AI CPUs



Source: BofA Global Research

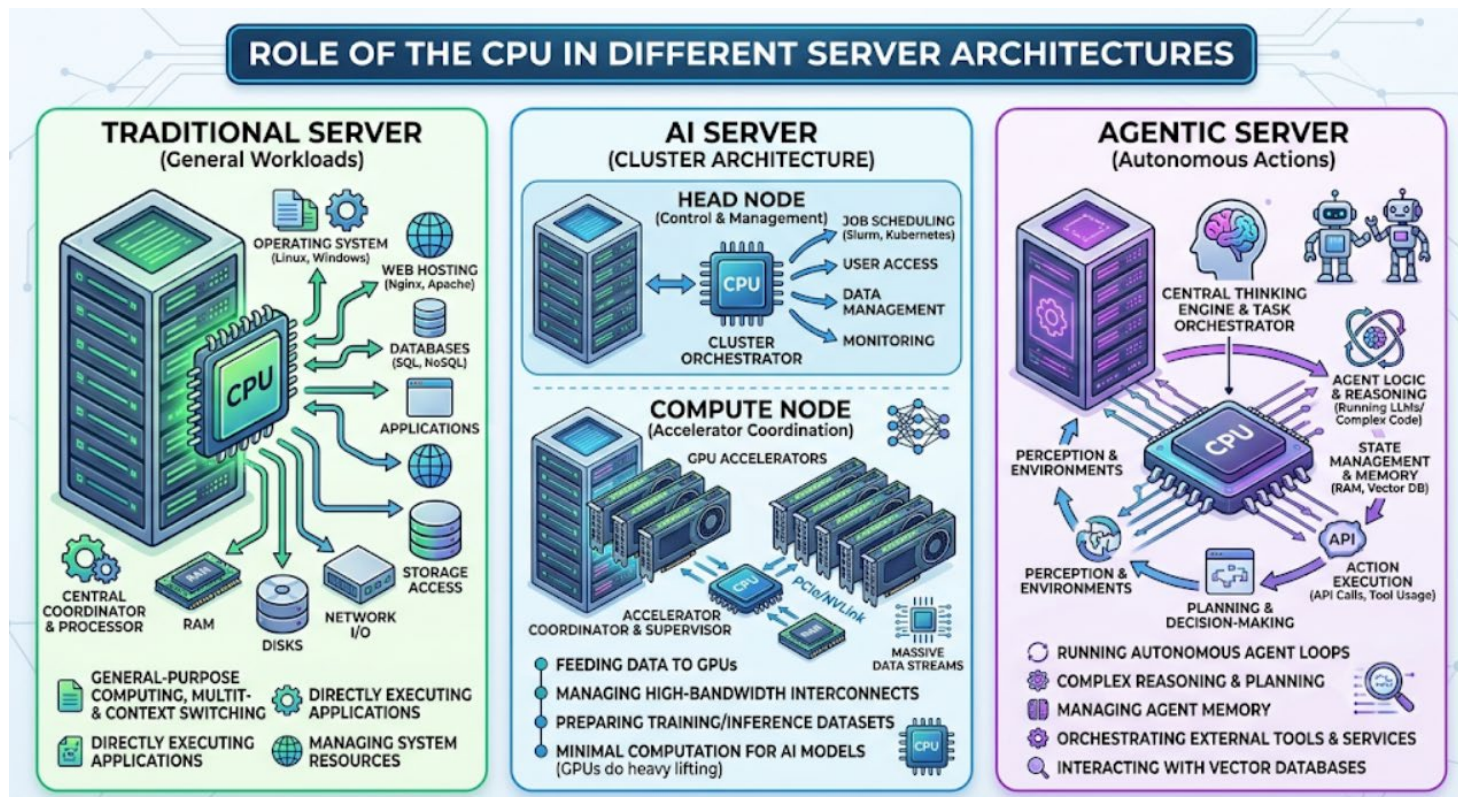
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The below infographic highlights how the role of the CPU evolves across different server architectures:

1. **Traditional Server:** The CPU is the primary "jack-of-all-trades," managing the OS, applications, and direct I/O for general workloads. (~\$30bn CY30E TAM)
2. **AI Server Cluster:** The CPU's role bifurcates into a Head Node for cluster orchestration and management, and a Compute Node focused on supervising and feeding data to GPUs. (~\$70bn CY30E TAM)
3. **Agentic Server:** The CPU becomes the central orchestrator for autonomous actions, managing the high-level reasoning loops, memory state, and tool integration required for AI agents. (~\$70bn CY30E TAM)

Exhibit 3: We flag three key roles of CPUs in servers: 1. Traditional, 2. AI compute/head node, 3. AI agentic node

Role of the CPU in different server architectures



Source: BofA Global Research

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\$170bn+ Server CPU TAM

Given the rising role of CPUs in AI head node (compute node) and agentic AI, we now expect server CPU TAM to reach ~\$170bn by CY30E from ~\$35bn in CY25 (+37% CAGR), versus prior ~\$125bn outlook (+29% CAGR).

Notably, we break out the market into 4 different vendor/types: **1) INTC, 2) AMD, 3) ARM (merchant), and 4) ARM (custom/ASIC).**

- For **ARM merchant CPUs**, we now assume ASPs that are ~1.25x higher than average AMD CPUs, given ARM merchant almost only caters to hyperscale/AI applications, versus AMD mostly hyperscale with a bit of enterprise exposure.
- For **ARM custom CPUs**, we now assume ASPs that are ~75% of typical AMD solutions, given continued mix shift toward hyperscale solutions (higher ASPs), but merchant server CPUs typically carry 40-60% GM (i.e. 40-60% is COGS or the cost that AWS/Google would theoretically be paying for their custom CPUs)

Exhibit 4: We see total server CPU TAM reaching \$170bn+ by CY30 from \$35bn in CY25, growing at +37% CAGR

Server CPU TAM (by revenue, units, ASP)

Server CPU TAM	2023	2024	2025	2026E	2027E	2028E	2029E	2030E	CY25-30 CAGR
Revenue (\$bn)									
Total	\$19.5	\$22.5	\$35.2	\$59.6	\$89.8	\$110.4	\$138.2	\$170.1	37.1%
INTC	\$13.2	\$13.6	\$14.4	\$19.8	\$24.7	\$29.6	\$35.1	\$41.2	23.4%
AMD	\$5.6	\$7.1	\$9.6	\$16.4	\$22.2	\$28.6	\$35.7	\$43.6	35.4%
ARM	\$0.7	\$1.8	\$11.2	\$23.4	\$42.9	\$52.2	\$67.5	\$85.4	50.0%
ARM Merchant	\$0.0	\$0.7	\$10.3	\$19.5	\$34.8	\$38.2	\$47.9	\$60.6	42.6%
NVDA		\$0.7	\$10.3	\$19.5	\$33.7	\$35.4	\$39.0	\$42.2	32.7%
ARM					\$0.7	\$1.8	\$5.9	\$12.3	
QCOM					\$0.4	\$0.9	\$3.0	\$6.1	
ARM Custom (Graviton, Axion, Cobalt, etc.)	\$0.7	\$1.0	\$1.0	\$3.9	\$8.1	\$14.1	\$19.6	\$24.8	91.0%
Units (mn)									
Total	24.5	24.2	29.2	39.3	49.0	57.1	66.0	74.7	20.7%
INTC	18.5	16.9	17.9	21.1	23.2	24.9	26.7	28.6	9.7%
AMD	4.8	5.5	7.2	10.4	12.5	14.5	16.3	18.1	20.2%
ARM	1.2	1.8	4.0	7.9	13.3	17.7	22.9	28.0	47.4%
ARM Merchant	0.0	0.2	2.6	4.6	7.2	8.2	10.9	14.2	40.9%
NVDA	0.0	0.2	2.6	4.6	6.7	7.1	7.7	8.1	25.9%
ARM	0.0	0.0	0.0	0.0	0.3	0.7	2.2	4.1	
QCOM	0.0	0.0	0.0	0.0	0.2	0.4	1.1	2.0	
ARM Custom (Graviton, Axion, Cobalt, etc.)	1.2	1.6	1.5	3.3	6.1	9.5	12.0	13.8	56.6%
ASP (\$)									
Total	\$797	\$931	\$1,204	\$1,517	\$1,833	\$1,934	\$2,095	\$2,277	13.6%
INTC	\$716	\$804	\$801	\$941	\$1,065	\$1,186	\$1,311	\$1,442	12.5%
AMD	\$1,164	\$1,299	\$1,322	\$1,581	\$1,778	\$1,976	\$2,183	\$2,401	12.7%
ARM	\$589	\$998	\$2,790	\$2,972	\$3,225	\$2,956	\$2,950	\$3,049	1.8%
ARM Merchant		\$4,000	\$4,000	\$4,240	\$4,820	\$4,658	\$4,393	\$4,255	1.2%
NVDA		\$4,000	\$4,000	\$4,240	\$5,000	\$5,000	\$5,100	\$5,202	5.4%
ARM					\$2,249	\$2,476	\$2,729	\$3,002	
QCOM					\$2,249	\$2,476	\$2,729	\$3,002	
ARM Custom (Graviton, Axion, Cobalt, etc.)	\$589	\$649	\$666	\$1,191	\$1,336	\$1,484	\$1,637	\$1,801	22.0%
Value Share (%)									
Total	100%	100%	100%	100%	100%	100%	100%	100%	
INTC	68%	61%	41%	33%	28%	27%	25%	24%	
AMD	29%	32%	27%	28%	25%	26%	26%	26%	
ARM	4%	8%	32%	39%	48%	47%	49%	50%	
ARM Merchant	0%	3%	29%	33%	39%	35%	35%	36%	
NVDA		3%	29%	33%	38%	32%	28%	25%	
ARM					1%	2%	4%	7%	
QCOM					0%	1%	2%	4%	
ARM Custom (Graviton, Axion, Cobalt, etc.)	4%	5%	3%	7%	9%	13%	14%	15%	

Source: BofA Global Research estimates, Mercury Research, IDC

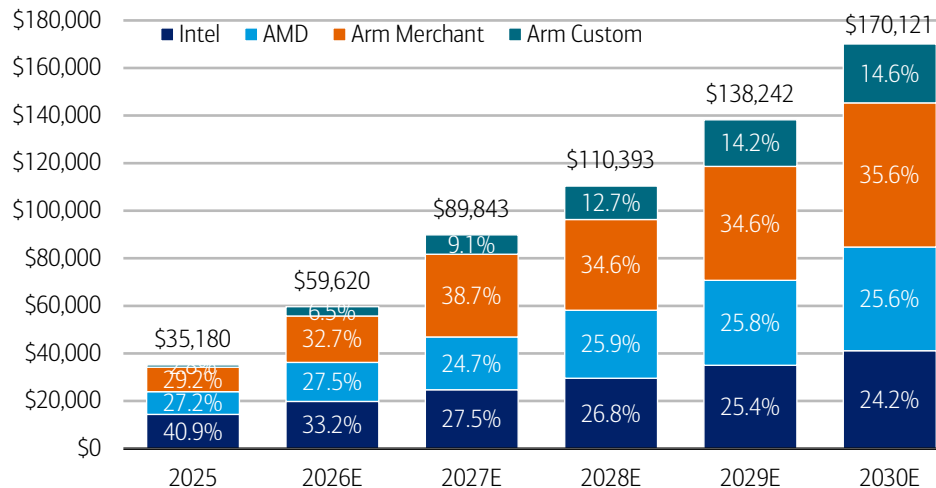


Market share outlook analysis

From a market share perspective, we continue to expect ARM to be the fastest gainer through CY30 as multiple new merchant (i.e. ARM AGI, NVDA Vera, QCOM CPU) and custom (i.e., AWS Graviton 5, Google Axion, Microsoft Cobalt) programs ramp.

Exhibit 5: We expect INTC/AMD to each represent ~25% value share, ARM merchant ~35%, ARM custom ~15% by CY30

Server CPU TAM (\$mn) and Value Market Share Outlook



Source: BofA Global Research estimates, Mercury Research, IDC

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- **INTC (~25% value share):** We see share declining to just ~24% of total server CPU value by CY30 from ~41% in CY25, though we expect INTC's continued relative strength in traditional enterprise workloads.

However, we see continued \$ revenue growth of +20% CAGR between CY26-30, particularly as INTC navigates through the industry-wide supply shortages, benefits from ASP expansion. By CY28, next-next-gen Coral Rapids should ramp and could begin to narrow the performance gap vs. competition.

- **AMD (~25% value share):** We see share of ~25-27% generally through CY30, given AMD's continued performance lead at the high-end (high-frequency, high-core-count AI head node and agentic node), offset by relative stronger ramps of ARM-based processors launching beginning CY27E.

Particularly, we flag AMD's leading core count portfolio, given higher core density within a fixed power envelope increases overall aggregate service throughput. 6th gen EPYC Venice will top out at 256 cores, while the current 5th gen Turin tops out at 192 cores, vs. NVDA Vera at 88 cores and INTC's current Granite Rapids topping at 128 cores and upcoming Diamond Rapids likely maxing at 192 cores.

Similar to INTC, we expect AMD to continue benefiting from the x86 ecosystem's advantages in security and RAS features, particularly important in agentic AI workloads.

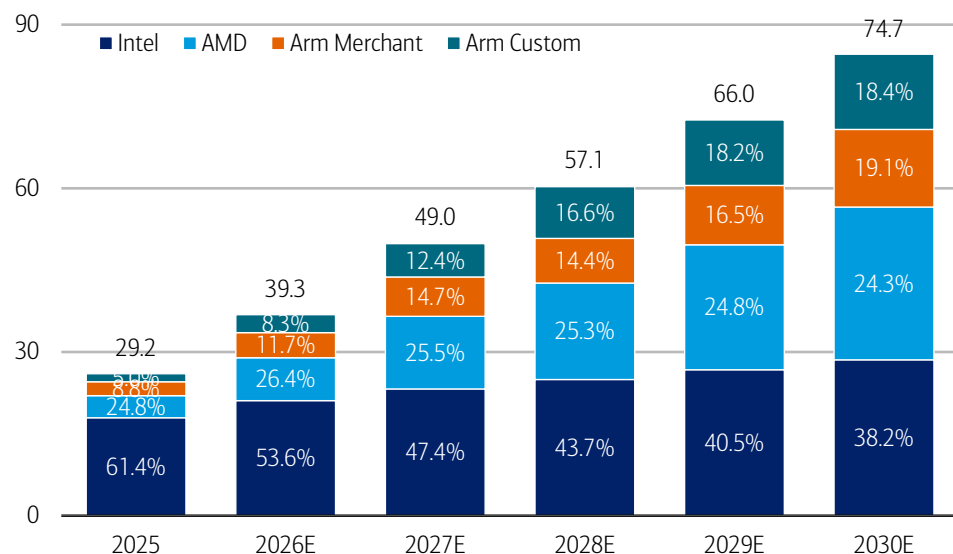
- **ARM (merchant; ~35% value share):** We see share rising to ~36% by CY30 from ~29% in CY25, led by a continued ramp of NVDA CPUs (in both compute/agentic nodes), as well as new ARM/QCOM merchant solutions ramping in CY27-28.
- **ARM (custom; ~15% value share):** We see significant share gain toward 15% by CY30 from <5% in CY23-25 as new custom ARM-based CPU solutions launch and ramp at multiple hyperscalers. Every year, the ARM content/ASP grows as core counts and performance per core grow.



On a unit share basis, we expect INTC to remain the largest at ~38% share by CY30.

Exhibit 6: We expect INTC to represent ~38% unit share, AMD ~24%, ARM merchant ~19%, and ARM custom ~18% by CY30

Server CPU TAM (mn units) and Unit Market Share Outlook



Source: BofA Global Research estimates, Mercury Research, IDC

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Traditional vs. AI Head Node (Compute) vs. Agentic Rack

In the following Exhibit, we also present the \$170bn CY30 server CPU TAM in 3 different key workloads: **1) traditional/infrastructure-as-a-service**, **2) AI compute node or head node**, and **3) AI agentic standalone CPU node**.

Particularly, we expect the two AI applications to represent ~50%/50% of the total CPU demand in AI data centers (~\$140bn) by CY30, as NVDA recently highlighted.

- **Traditional/laaS:** This includes the most traditional and conventional CPU use cases for enterprises on-prem and on public cloud.

This segment represented the vast majority of the server CPU market in 2010-2020, growing ~11-12% CAGR during that period. For the next five years, we expect this growth rate to continue, at ~11-12% sales CAGR range, or units growing ~7-9% CAGR, with ASPs growing ~3-5% CAGR.

- **AI compute node, head node:** This includes CPUs that are attached to GPUs/XPUs and are acting as supporting roles within the compute rack (most pronouncedly NVDA's NVL72 rack, AMD's MI455 'Helios' rack, TPUv8 racks). In other words, these CPUs are used as part of the scale-up domain, facilitating XPU compute.

While the vast majority of initial AI CPU use cases (~2022-2026) were compute/head nodes, we anticipate a mix shift toward agentic AI standalone CPU racks over time (2026-2030), eventually the two AI use cases likely representing ~50/50 of all AI workloads.

- **AI agentic standalone CPU node/rack:** This includes the new CPU-only standalone racks, such as the NVDA Vera CPX rack which is part of the overall 40-rack Vera Rubin SuperPOD. These CPU-only racks are highly optimized for agentic AI workloads which require sequential processing in tandem with GPU/XPU's parallel compute processing, and are responsible for orchestration, memory control, I/O, etc.

Exhibit 7: Of the \$170bn CY30 CPU TAM, we see ~\$30bn for traditional/laaS, ~\$70bn for AI compute/head node, ~\$70bn for agentic AI CPU racks
 Server CPU TAM Breakout by Workload

		2022	2023	2024	2025	2026	2027	2028	2029E	2030E	CAGR '20-'25	CAGR '25-'30	CAGR '10-'20	Notes
CPU Sales (\$mn)	Total	\$23.9	\$19.5	\$22.5	\$35.2	\$59.6	\$89.8	\$110.4	\$138.2	\$170.1	7.3%	37.1%		Bottom-Up Build
	Intel	\$18.1	\$13.2	\$13.6	\$14.4	\$19.8	\$24.7	\$29.6	\$35.1	\$41.2	-9.0%	23.4%		
	AMD	\$5.6	\$5.6	\$7.1	\$9.6	\$16.4	\$22.2	\$28.6	\$35.7	\$43.6	44.0%	35.4%		
	Arm	\$0.2	\$0.7	\$1.8	\$11.2	\$23.4	\$42.9	\$52.2	\$67.5	\$85.4	167.8%	50.0%		
	Traditional/laaS	\$22.9	\$18.0	\$17.4	\$17.0	\$19.0	\$21.2	\$23.7	\$26.4	\$29.5	-6.9%	11.7%	11.9%	
	Intel	\$17.3	\$12.2	\$11.3	\$10.7	\$12.0	\$13.5	\$15.2	\$17.1	\$19.2	-14.0%	12.4%	11.9%	
	AMD	\$5.4	\$5.1	\$5.2	\$5.3	\$5.8	\$6.4	\$7.0	\$7.6	\$8.4	27.8%	9.8%	10.9%	
	Arm	\$0.2	\$0.7	\$0.9	\$1.0	\$1.2	\$1.3	\$1.5	\$1.7	\$1.9	66.4%	13.4%	N/A	
	AI (Compute/Agentic)	\$1.0	\$1.5	\$5.1	\$18.2	\$40.6	\$68.6	\$86.7	\$111.8	\$140.6	123.3%	50.6%		
	Intel	\$0.8	\$1.0	\$2.4	\$3.7	\$7.8	\$11.2	\$14.4	\$18.0	\$22.0	62.4%	43.2%		
	AMD	\$0.2	\$0.5	\$1.9	\$4.3	\$10.6	\$15.8	\$21.6	\$28.0	\$35.2	292.4%	52.1%		
	Arm	\$0.0	\$0.0	\$0.9	\$10.2	\$22.2	\$41.6	\$50.7	\$65.8	\$83.4		52.3%		
	Compute/Head Node	\$1.0	\$1.5	\$5.1	\$18.2	\$34.5	\$44.6	\$52.0	\$61.5	\$70.3	123.3%	31.1%		
	Intel	\$0.8	\$1.0	\$2.4	\$3.7	\$6.6	\$7.3	\$8.6	\$9.9	\$11.0	62.4%	24.7%		
	AMD	\$0.2	\$0.5	\$1.9	\$4.3	\$9.0	\$10.3	\$13.0	\$15.4	\$17.6	292.4%	32.4%		
	Arm	\$0.0	\$0.0	\$0.9	\$10.2	\$13.2	\$27.0	\$30.4	\$36.2	\$41.7		32.5%		
	Agentic AI Node	\$0.0	\$0.0	\$0.0	\$0.0	\$6.1	\$24.0	\$34.7	\$50.3	\$70.3				
	Intel	\$0.0	\$0.0	\$0.0	\$0.0	\$1.2	\$3.9	\$5.8	\$8.1	\$11.0				
	AMD	\$0.0	\$0.0	\$0.0	\$0.0	\$1.6	\$5.5	\$8.6	\$12.6	\$17.6				
	Arm	\$0.0	\$0.0	\$0.0	\$0.0	\$9.0	\$14.6	\$20.3	\$29.6	\$41.7				
CPU Units (mn)	Total	36.6	24.5	24.2	29.2	39.3	49.0	57.1	66.0	74.7	-3.9%	20.7%		Bottom-Up Build
	Intel	30.7	18.5	16.9	17.9	21.1	23.2	24.9	26.7	28.6	-11.6%	9.7%		
	AMD	5.4	4.8	5.5	7.2	10.4	12.5	14.5	16.3	18.1	27.2%	20.2%		
	Arm	0.4	1.2	1.8	4.0	7.9	13.3	17.7	22.9	28.0	77.1%	47.4%		
	Traditional/laaS	35.6	23.2	20.5	20.9	22.4	24.0	25.7	27.5	29.5	-9.9%	7.1%	9.1%	
	Intel	29.9	17.6	14.8	14.8	15.8	16.9	18.1	19.4	20.7	-14.8%	7.0%	9.3%	
	AMD	5.2	4.4	4.2	4.5	4.8	5.2	5.5	5.8	6.2	15.8%	6.6%	5.9%	
	Arm	0.4	1.2	1.5	1.6	1.8	2.0	2.1	2.3	2.5	48.2%	9.0%	N/A	
	AI (Compute/Agentic)	1.0	1.3	3.7	8.3	16.9	25.0	31.4	38.4	45.2	103.5%	40.5%		
	Intel	0.8	0.9	2.2	3.2	5.2	6.3	6.8	7.4	7.8	68.6%	19.9%		
	AMD	0.2	0.4	1.3	2.7	5.5	7.3	9.0	10.5	11.9	262.3%	34.4%		
	Arm	0.0	0.0	0.3	2.4	6.1	11.4	15.5	20.6	25.5	452.6%	60.6%		
	AI (Compute/Agentic)	1.0	1.2	\$1.4	\$2.2	\$2.4	\$2.7	\$2.8	\$2.9	\$3.1	9.7%	7.2%		
	Intel	\$1.0	\$1.1	\$1.1	\$1.2	\$1.5	\$1.8	\$2.1	\$2.4	\$2.8	-3.7%	19.4%		
	AMD	\$1.2	\$1.4	\$1.5	\$1.6	\$1.9	\$2.2	\$2.4	\$2.7	\$3.0	8.3%	13.2%		
	Arm			\$3.2	\$4.3	\$3.7	\$3.7	\$3.3	\$3.2	\$3.3		-5.2%		
CPU ASP (\$k)	Total	\$0.7	\$0.8	\$0.9	\$1.2	\$1.5	\$1.8	\$1.9	\$2.1	\$2.3	11.7%	13.6%		Bottom-Up Build
	Intel	\$0.6	\$0.7	\$0.8	\$0.8	\$0.9	\$1.1	\$1.2	\$1.3	\$1.4	2.9%	12.5%		
	AMD	\$1.0	\$1.2	\$1.3	\$1.3	\$1.6	\$1.8	\$2.0	\$2.2	\$2.4	13.2%	12.7%		
	Arm	\$0.5	\$0.6	\$1.0	\$2.8	\$3.0	\$3.2	\$3.0	\$3.0	\$3.0	51.2%	1.8%		
	Traditional/laaS	\$0.6	\$0.8	\$0.8	\$0.8	\$0.8	\$0.9	\$0.9	\$1.0	\$1.0	3.3%	4.3%	2.5%	
	Intel	\$0.6	\$0.7	\$0.8	\$0.7	\$0.8	\$0.8	\$0.8	\$0.9	\$0.9	1.0%	5.0%	2.4%	
	AMD	\$1.0	\$1.1	\$1.2	\$1.2	\$1.2	\$1.2	\$1.3	\$1.3	\$1.3	10.3%	3.0%	4.7%	
	Arm	\$0.5	\$0.6	\$0.6	\$0.6	\$0.7	\$0.7	\$0.7	\$0.7	\$0.8	12.3%	4.0%	N/A	
	AI (Compute/Agentic)	\$1.0	\$1.2	\$1.4	\$2.2	\$2.4	\$2.7	\$2.8	\$2.9	\$3.1	9.7%	7.2%		
	Intel	\$1.0	\$1.1	\$1.1	\$1.2	\$1.5	\$1.8	\$2.1	\$2.4	\$2.8	-3.7%	19.4%		
	AMD	\$1.2	\$1.4	\$1.5	\$1.6	\$1.9	\$2.2	\$2.4	\$2.7	\$3.0	8.3%	13.2%		
	Arm			\$3.2	\$4.3	\$3.7	\$3.7	\$3.3	\$3.2	\$3.3		-5.2%		
	AI (Compute/Agentic)	\$1.0	\$1.2	\$1.4	\$2.2	\$2.4	\$2.7	\$2.8	\$2.9	\$3.1	9.7%	7.2%		
	Intel	\$1.0	\$1.1	\$1.1	\$1.2	\$1.5	\$1.8	\$2.1	\$2.4	\$2.8	-3.7%	19.4%		
	AMD	\$1.2	\$1.4	\$1.5	\$1.6	\$1.9	\$2.2	\$2.4	\$2.7	\$3.0	8.3%	13.2%		
	Arm			\$3.2	\$4.3	\$3.7	\$3.7	\$3.3	\$3.2	\$3.3		-5.2%		
CPU-XPU Ratio	AI CPU Unit TAM	1.0	1.3	3.7	8.3	16.9	25.0	31.4	38.4	45.2	103.5%	40.5%		Bottom-Up Build AI Accelerator Model
	XPU Unit TAM	4.9	6.2	11.0	16.3	23.6	28.7	28.9	32.8	35.6	39.0%	16.9%		
	AI CPU/XPU Ratio (%)	19%	20%	33%	51%	71%	87%	108%	117%	127%				

Source: BofA Global Research estimates, Mercury Research, IDC

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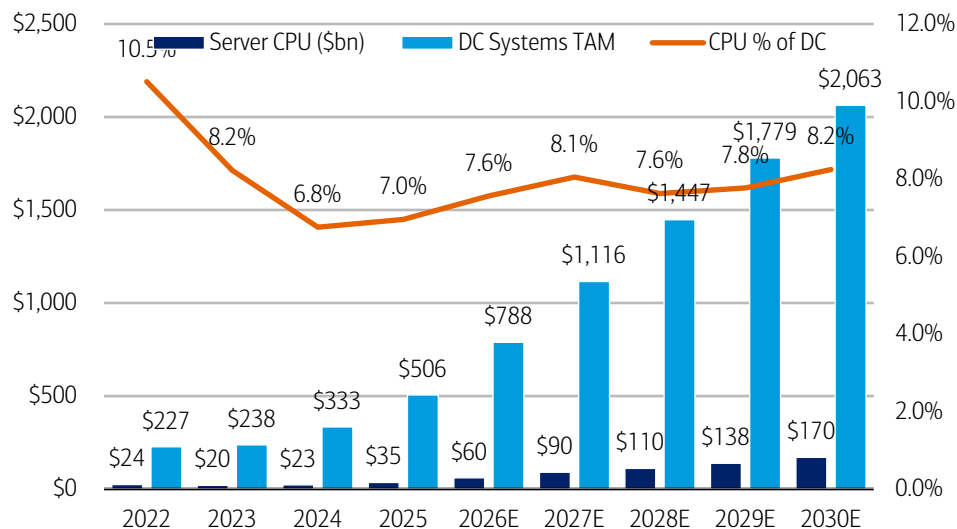
CPUs now 8%+ of Data Center Systems TAM

At ~\$170bn+ TAM, we now believe server CPUs to make up 8%+ of overall data center systems TAM over time (vs. prior 5-6%), up modestly from a low of <7% in 2025 when the value share of AI training was at its peak.

Over time, we expect the value of CPUs in DC systems to modestly expand given their roles in AI inference and agentic AI workloads, though still at a more limited 7-8% range given other components (such as AI accelerators, HBM memory, networking, connectivity) also remain critical to the overall system.

Exhibit 8: We expect server CPUs to make up 8%+ of total DC TAM value over time, modestly expanding over time as their roles increase in agentic AI workloads

Server CPU as a % of DC Systems TAM



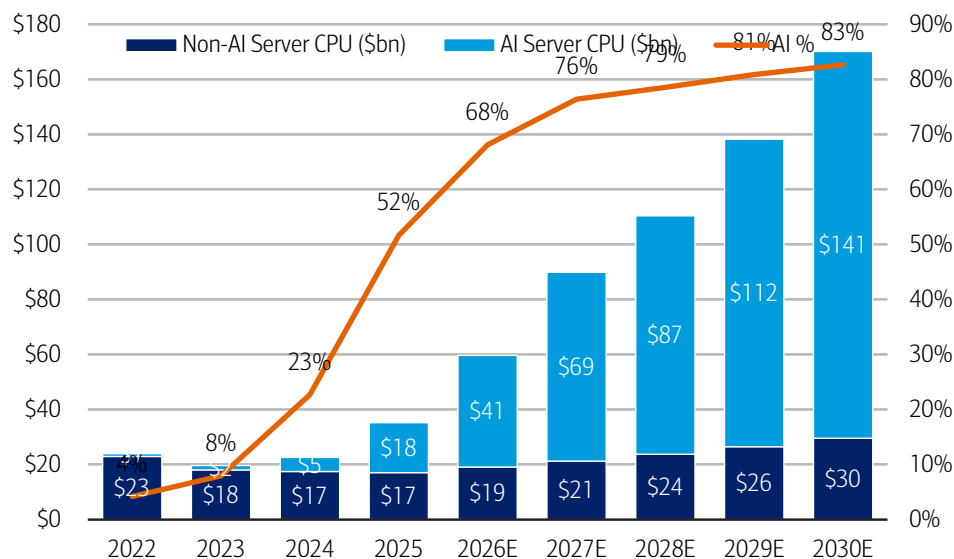
Source: BofA Global Research estimates, Mercury Research, IDC, LightCounting, Gartner

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For AI specifically, we expect AI CPUs to represent 80%+ of total server CPUs over time.

Exhibit 9: Server CPU TAM could reach \$170bn+ by CY30E, with AI CPUs representing 80%+

Server CPU TAM and AI Server CPU as a % of mix



Source: BofA Global Research estimates, Mercury Research, IDC

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AMD: Reiterate Buy, raise PO to \$560 from \$500

We flag increased server CPU market demand visibility, view AMD as the best positioned CPU vendor (unit/ASP expansion), raise estimates accordingly for CY26-28, and also raise our PO to \$560 from \$500 on unchanged 42x CY27 PE.

AMD CPU Advantages

As we mentioned in above sections, we highlight AMD's leading CPU portfolio in both **high frequency** and **core counts**.

High frequency (5GHz+)

For head node (compute node)-type workloads, CPUs with higher single-threaded performance (high-frequency) are key to overall rack performance.

NVDA's Grace (1H23) and Vera (2H26) CPUs were developed specifically for this. While NVDA provides no public disclosure of CPU frequency, third-party benchmarks for Grace point to ~3.35GHz peak perf for the 3-year old CPU. While slower than AMD Turin's (4Q24) peak 5.0GHz and INTC Granite Rapids' (3Q24) peak 4.3GHz, NVDA CPUs also benefit from the tight co-design and co-integration of CPUs and GPUs via fast NVLink C2C connect, as opposed to slower PCIe connectivity for most other CPU vendors.

Meanwhile, AMD's Turin sports up to 5.0GHz peak per-core performance with high-frequency (HF) SKUs, while also utilizing its own InfinityFabric interconnect (for GPU-to-GPU, GPU-to-CPU) in scale-up.

Once the upcoming Venice CPUs launch in 2H26, we expect this single-threaded performance lead to extend, making AMD CPUs ideal for head/compute node in AI scale-up clusters outside of the NVDA ecosystem.

Exhibit 10: AMD's Turin leads in peak frequency at 5.0GHz, with the upcoming Venice expected to lead max core count at 256-core/512-thread

Major CPU Specs, including max # of cores/threads and peak frequency

Vendor	INTC	INTC	AMD	AMD	NVDA	NVDA	ARM	QCOM	AWS	GOOGL	MSFT
Architecture	x86	x86	x86	x86	ARM	ARM	ARM	ARM	ARM	ARM	ARM
Merchant/Custom	Merchant	Merchant	Merchant	Merchant	Merchant	Merchant	Merchant	Merchant	Custom	Custom	Custom
Product	Granite Rapids	Diamond Rapids	Turin	Venice	Grace	Vera	AGI	TBD	Graviton 5	Axion	Cobalt
Process	Intel 3	Intel 18A	TSMC 4nm	TSMC 2nm	TSMC 4nm	TSMC 3nm	TSMC 3nm	TBD	TSMC 3nm	-	-
Launch	3Q24	3Q26	4Q24	3Q26	2Q23	3Q26	1Q27	2027E	2Q26	4Q24	4Q24
Peak Frequency	3.9GHz	TBD	5.0GHz	TBD	3.35GHz	TBD	3.7GHz	TBD	-	3.0GHz?	3.4GHz
Max Cores/Threads	128C/256T	192C	192C/384T	256C/512T	74C	88C/176T	136C	TBD	192C	-	-
Multi-threading	Hyper	N/A	Simultaneous	Simultaneous	N/A	Spatial	N/A	TBD	N/A	-	-

Source: BofA Global Research estimates

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Core counts (up to 256 cores vs. 88-192 competition)

For agentic AI workloads and standalone CPU nodes/racks, CPUs with higher number of cores (and threads via multi-threading) are key to the overall system performance.

This is because within a fixed power envelope of a server/rack, higher core density increases overall aggregate service throughput.

AMD's upcoming 6th gen EPYC Venice (2H26) is expected to top out at 256 cores, while the current 5th gen Turin tops out at 192 cores. This is much higher than NVDA's upcoming Vera (2H26) at 88 cores, as well as INTC's current Granite Rapids topping at 128 cores and upcoming Diamond Rapids (~2H26) likely maxing at 192 cores.

As a result, AMD EPYC CPUs generally lead modeled rack-level results over others, with the EPYC 9965 ("Turin," 192C) delivering a 2.37x normalized geometric mean advantage over NVDA Vera (88 Olympus cores). This lead is expected to extend to 3.30x Vera with the EPYC "Venice" (256C). For context, INTC's Xeon 6980P ("Granite Rapids-AP," 128C) also turns in 1.46x over NVDA Vera in high-core-count agentic workloads.



Upcoming catalyst: Advancing AI 2026 (July)

Separately, we flag AMD's upcoming "Advancing AI 2026" Event in San Francisco, with CEO Dr. Lisa Su's keynote scheduled on July 23.

We anticipate AMD to formally launch its upcoming "Venice" and "Verano" CPUs, the "MI455X" GPU, as well as the "Helios" rack for scale-up compute.

- **CPUs:** We expect a great focus on the performance/core-count lead of AMD CPUs, versus not only INTC but also many other new/existing ARM-based competitors including NVDA, ARM, QCOM who have initially designed CPUs for head/compute-node purposes, not necessarily the agentic AI node/rack purposes (which benefit from having many cores/multi-threading).

We also expect AMD to play up its advantage in the x86 ecosystem with security and RAS features, particularly important in agentic AI workloads.

- **GPUs:** We flag potential for AMD to announce additional GW-scale GPU customers (as full MI455X-based "Helios" customers), on top of the current existing OpenAI and Meta lineup. For reference, AMD has been in engagement with many hyperscale customers to date.

Valuation

Our new PO of \$560 is based on an unchanged 42x CY27E PE, toward middle/upper range of historical 13x-58x, but is well supported by AMD's 50%+ annual EPS CAGR potential and its AI CPU/GPU share gain potential.

Exhibit 11: We raise AMD CY/FY27-28 sales by 6-10%, pf-EPS by 10-16% on increased CPU market outlook and potential new GW-scale GPU engagements

AMD Estimate Changes

	Non-GAAP Sales (\$mn)			EPS (Non-GAAP, ex-SBC)		
	OLD	NEW	delta	OLD	NEW	delta
1Q26	\$13,577	\$13,577	\$0	\$0.29	\$0.29	\$0.00
2Q26E	\$14,302	\$14,302	\$0	\$0.20	\$0.20	\$0.00
3Q26E	\$14,895	\$14,895	\$0	\$0.25	\$0.25	\$0.01
4Q26E	\$15,451	\$15,478	\$27	\$0.30	\$0.31	\$0.01
2026E	\$58,225	\$58,252	\$27	\$1.04	\$1.06	\$0.02
YoY%	10.2%	10.2%	0%	143.6%	148.0%	2%
1Q27E	\$15,582	\$15,664	\$82	\$0.32	\$0.34	\$0.02
2Q27E	\$16,031	\$16,170	\$139	\$0.34	\$0.37	\$0.03
3Q27E	\$16,909	\$17,082	\$173	\$0.43	\$0.47	\$0.04
4Q27E	\$17,718	\$18,018	\$300	\$0.49	\$0.54	\$0.05
2027E	\$66,240	\$66,933	\$694	\$1.58	\$1.72	\$0.14
YoY%	13.8%	14.9%	1%	52.0%	63.0%	9%
1Q28E	\$17,659	\$18,029	\$370	\$0.49	\$0.55	\$0.06
2Q28E	\$18,116	\$18,572	\$456	\$0.52	\$0.59	\$0.07
3Q28E	\$19,085	\$19,638	\$553	\$0.59	\$0.67	\$0.07
4Q28E	\$19,962	\$20,647	\$686	\$0.64	\$0.73	\$0.08
2028E	\$74,822	\$76,887	\$2,065	\$2.25	\$2.53	\$0.28
YoY%	13.0%	14.9%	3%	42.7%	47.0%	12%

Source: BofA Global Research estimates

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INTC: Upgrade to Buy from U/P, PO \$135

We double-upgrade INTC to Buy from Underperform, raise ests/PO to \$135 from \$96.

We highlight INTC's expanded opportunity in server CPUs (similar to AMD described in the AMD section) which it can manufacture itself (gives higher supply visibility), as well as increased opportunities in its external foundry business.

Fully established IDM by CY30, EPS power \$6+

Particularly, we now value INTC as a fully integrated IDM company with EPS power of ~\$6.24 by CY30, with our new PO based on 25x CY30E PE, discounted back two years.

This is a change from our prior CY28 sum-of-parts valuation methodology (more in Exhibit 13) which we now believe under-represents many of the company's CPU and foundry potentials that are further out in CY29-30+.

- **Products:** We now expect INTC server CPUs sales to reach ~\$40bn+ by CY30E (vast majority of DCAI segment is server CPUs), or about ~25% share of \$170bn TAM.
- **Foundry:** While we are not changing our base case foundry segment model until there is further clarity on specific deal details, we present the multiple deals INTC is currently engaged with, size the potential opportunities below. Opportunities span across: Apple M-Series wafers, MediaTek TPU wafers, Terafab IP/packaging, and potentially other ARM-based server CPUs, edge AI expansion into Client – all made more possible via recent CDNS 14A collab announcement.

Exhibit 12: We expect \$6+ INTC EPS power by CY30, now see total company value of \$135, versus \$96 prior based on CY28 sum-of-parts

INTC Sales Power, EPS Power, Valuation Methodology

	CY26E	CY27E	CY28E	CY29E	CY30E	CY30E Methodology	CAGR
Products	\$55.0	\$61.4	\$68.4	\$77.6	\$86.1		11.9%
CCG	\$31.6	\$32.8	\$34.4	\$38.6	\$42.4		
DCAI	\$23.4	\$28.6	\$33.9	\$39.0	\$43.7	~25% Share of \$170bn TAM	
Foundry	\$1.1	\$8.0	\$17.5	\$30.8	\$47.1		158.4%
A/P	\$0.4	\$1.9	\$3.5	\$4.8	\$6.0	30.0% Share of \$20bn 2.5D/3D TAM	
Wafer	\$0.5	\$5.7	\$11.4	\$17.0	\$23.1	~6-7% Share of \$380bn Wafer Foundry TAM	
Apple	\$0.0	\$2.5	\$5.4	\$8.0	\$10.6	40.0% Share of M-Series, 25% A-Series	
TPU/MediaTek	\$0.5	\$3.3	\$6.0	\$9.0	\$12.4	100.0% Share of TPU (MediaTek)	
Terafab	\$0.1	\$0.3	\$1.5	\$4.0	\$8.0	IP, Packaging, Wafer (partial): mid-2028 begin	
Other HPC Products	\$0.0	\$0.0	\$1.0	\$5.0	\$10.0	ARM-based server CPU opp'ty	
Other	\$2.3	\$2.5	\$2.5	\$2.6	\$2.8		
Net Sales	\$58.4	\$71.9	\$88.3	\$111.0	\$135.9		23.5%
Products EBIT	\$18.0	\$21.6	\$25.7	\$30.8	\$33.2	38.5% vs. OpM Target (38.5%)	
Foundry EBIT	-\$9.3	-\$6.6	-\$1.8	\$3.1	\$8.5	18.0% vs. OpM Target (27.5%)	
Other, Unallocated	-\$1.1	-\$1.0	-\$1.0	-\$1.0	-\$1.0		
Total EBIT (non-GAAP)	\$7.6	\$13.9	\$22.8	\$32.9	\$40.6		52.0%
OpM (%)	13.0%	19.4%	25.8%	29.6%	29.9%		
Interest Income	-\$0.2	-\$0.8	-\$0.8	-\$0.8	-\$0.8		
EBT	\$7.4	\$13.2	\$22.0	\$32.1	\$39.8		
Tax Rate	11.5%	12.0%	12.0%	12.0%	12.0%		
Net Income	\$6.5	\$11.6	\$19.4	\$28.3	\$35.1		
NCI Adj.	-\$1.1	-\$1.2	-\$1.2	-\$1.2	-\$1.3		
Net Income (non-GAAP)	\$5.4	\$10.4	\$18.2	\$27.0	\$33.8		
EPS Power	\$1.06	\$2.01	\$3.47	\$5.07	\$6.24		55.9%
Shares Outstanding	5,106	5,172	5,248	5,326	5,406		
Multiple					25x		
Price Objective (25x CY30E discounted back)			\$135	\$145	\$156		
Discount			-7%	-7%			

Source: BofA Global Research estimates



Exhibit 13: We now see ~\$3.47 CY28E EPS Power vs. ~\$2.25 prior, on greater server CPU/Foundry opportunities

INTC Valuation Methodology: New vs. Old

Prior Method	CY28E New	CY28E Old			
	Total	Total	IDM	Foundry	Apple
Products	\$68.4	\$66.3	\$66.3	\$0.0	\$0.0
CCG	\$34.4	\$34.4	\$34.4		
DCAI	\$33.9	\$31.9	\$31.9		
Foundry	\$17.5	\$11.4	\$0.0	\$6.0	\$5.4
A/P	\$3.5	\$3.5		\$3.5	
Wafer	\$11.4	\$7.9	\$0.0	\$2.5	\$5.4
Apple	\$5.4	\$5.4			\$5.4
Other Wafer	\$6.0	\$2.5		\$2.5	
Terafab	\$1.5	\$0.0			
Other HPC Products	\$1.0	\$0.0			
Other	\$2.5	\$2.5	\$2.5		
Net Sales	\$88.3	\$80.2	\$68.8	\$6.0	\$5.4
Products EBIT	\$25.7	\$15.1	\$15.1		
Foundry EBIT	-\$1.8	\$1.5		\$1.5	
Other, Unallocated	-\$1.0	-\$1.0	-\$1.0		
Total EBIT (non-GAAP)	\$22.8	\$15.6	\$14.1	\$1.5	\$0.0
OpM (%)	25.8%	45.4%	20.4%	25.0%	0.0%
Interest Income	-\$0.8	-\$0.8	-\$0.8		
EBT	\$22.0	\$14.8	\$13.3	\$1.5	\$0.0
Tax Rate	12.0%	12.0%	12.0%	12.0%	12.0%
Net Income	\$19.4	\$13.0	\$11.7	\$1.3	\$0.0
NCI Adj.	-\$1.2	-\$1.2	-\$1.2		
Net Income (non-GAAP)	\$18.2	\$11.8	\$10.5	\$1.3	\$0.0
EPS Power	\$3.47	\$2.25	\$2.00	\$0.25	\$0.00
Shares Outst.	5,248	5,246	5,246	5,246	5,246
Methodology		SOP	PE	EV/S	EV/S
Multiple		43x	37x	10x	10x
Price Power	\$135	\$96	\$74	\$11	\$10

Source: BofA Global Research estimates

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Exhibit 14: We raise INTC CY27-28E sales by 1-3%, pf-EPS by 9-12%

INTC Estimate Changes

Non-GAAP Sales (\$mn)				EPS (Non-GAAP, ex-SBC)		
	OLD	NEW	delta	OLD	NEW	delta
2026E	\$58,225	\$58,252	\$27	\$1.04	\$1.06	\$0.02
YoY%	10.2%	10.2%	0%	143.6%	148.0%	2%
1Q27E	\$15,582	\$15,664	\$82	\$0.32	\$0.34	\$0.02
2Q27E	\$16,031	\$16,170	\$139	\$0.34	\$0.37	\$0.03
3Q27E	\$16,909	\$17,082	\$173	\$0.43	\$0.47	\$0.04
4Q27E	\$17,718	\$18,018	\$300	\$0.49	\$0.54	\$0.05
2027E	\$66,240	\$66,933	\$694	\$1.58	\$1.72	\$0.14
YoY%	13.8%	14.9%	1%	52.0%	63.0%	9%
1Q28E	\$17,659	\$18,029	\$370	\$0.49	\$0.55	\$0.06
2Q28E	\$18,116	\$18,572	\$456	\$0.52	\$0.59	\$0.07
3Q28E	\$19,085	\$19,638	\$553	\$0.59	\$0.67	\$0.07
4Q28E	\$19,962	\$20,647	\$686	\$0.64	\$0.73	\$0.08
2028E	\$74,822	\$76,887	\$2,065	\$2.25	\$2.53	\$0.28
YoY%	13.0%	14.9%	3%	42.7%	47.0%	12%

Source: BofA Global Research estimates

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ARM: Reit. Neutral, raise PO to \$335

We see ARM as one of the most prominent beneficiaries of the rising server CPU tide, as the fastest overall ecosystem share gainer through CY30.

As we've highlighted in above sections, we see ARM's collective server CPU value share to reach ~50% by CY30 across: ~35% in merchant CPUs (NVDA, ARM, QCOM, etc.) and ~15% in custom CPUs (hyperscale ASIC such as AWS Graviton, GOOGL Axion, MSFT Cobalt).

However, at ~\$300+, we view the stock as fairly valued, with our new sum-of-parts valuation methodology yielding a PO of \$335, versus our previous total company valuation of \$245 based on 69x CY28E PE.

Sum-of-Parts Valuation: IP & Chip Businesses

We flag the immense opportunities ahead in standalone chip business (vs. historical only IP business), and we now value the company on a sum-of-parts basis.

- **IP business:** For the historical IP business, we continue to value based on ~2.0x PEG framework, in-line with high-growth semis and IP peers.

This is based on +20% Royalty sales CAGR and +12% Licensing sales CAGR, which at 65% combined OpM would yield EPS power of \$5+ by FY31, or +25% EPS CAGR.
- **Chip business:** For the new chiplet business, we value based on ~35x PE, in-line with high-growth AI compute vendor average of ~35x.

We see potential for \$15bn chip sales power by FY31 (~CY30), in-line with ARM mgmt's outlook of 15% share of ~\$100bn CPU TAM. We use 30% OpM target that ARM outlined (in-line with ~30% OpM for more established vendors such as AMD).
- **Total company:** For the total company valuation, we discount back two years and combine – 1) **\$229 worth of IP business**, and 2) **\$106 worth of chip business** – totaling **~\$335 of total company value**.

Exhibit 15: We see total company value of \$335, versus \$245 prior, now based on CY30 sum-of-parts for IP and Chip businesses, discounted back 2 years

ARM Sales Power, EPS Power, Valuation Methodology

	FY26	FY31E - IP Only		FY31E - Chip Only		FY31E - Total Company
IP Sales	\$4,920	\$10,569				\$10,569
Royalty Sales	\$2,613	\$6,503	20% CAGR			
Licensing Sales	\$2,307	\$4,066	12% CAGR			
Chip Sales	\$0			\$15,000		\$15,000
CPU TAM				\$100,000	15% TAM & Share	
Total Sales Power	\$4,920	\$10,569	17% CAGR	\$15,000	25% CAGR	\$25,569 39% CAGR
IP EBIT	\$2,115	\$6,870	65% OpM			\$6,870 65% OpM
Chip EBIT	\$0			\$4,500	30% OpM	\$4,500 30% OpM
Total EBIT	\$2,115	\$6,870		\$4,500		\$11,370
Interest Income	\$107	\$50		\$50		\$100
EBT	\$2,222	\$6,920		\$4,550		\$11,470
Tax Rate	15.0%	15.0%		15.0%		15.0%
Net Income	\$1,889	\$5,882		\$3,868		\$9,749
EPS Power	\$1.77	\$5.35	25% CAGR	\$3.52	15% CAGR	\$8.86 38% CAGR
Shares Outst.	1,068	1,100		1,100		1,100
~CY30 Valuation		\$265	2.0x PEG	\$123	35x PE	\$388 44x Implied PE
~CY29 Valuation		\$246		\$114		\$361 -7% Discount
~CY28 Valuation		\$229		\$106		\$335 -7% Discount

Source: BofA Global Research estimates

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NVDA: Maintain Buy, top AI sector pick

Rising content, increasing opp'ty, solid supply visibility

Following 'GTC Taipei' CEO keynote on Jun-4 (see [our takeaways](#)), we hosted NVDA CFO Colette Kress for a well-attended keynote, and Sr. Director, IR Stewart Stecker for a bus tour investor meeting.

Key takeaways: **1)** Vera CPU opportunity is roughly split 50/50 between head node (compute rack) and agentic AI (standalone CPU rack), **2)** NVDA's \$200bn CPU TAM only includes agentic AI, RL-type workloads, not general purpose traditional CPU workloads, **3)** NVDA content to rise meaningfully every system gen, at ~\$40bn/GW for Blackwell, ~\$60-80bn/GW for Rubin/Rubin Ultra, and potentially ~\$100bn/GW for Feynman, as NVDA consistently adds new chips/racks to the system; **4)** NVDA's full-stack architecture and ecosystem incumbency provides advantages in both hyperscale and ACIE; **5)** GPU useful life may be longer than depreciable life, as 3-4 year old Hoppers are still fully utilized; and **6)** NVDA's \$124bn purchase commitment suggests robust demand/supply visibility ahead for AI infra.

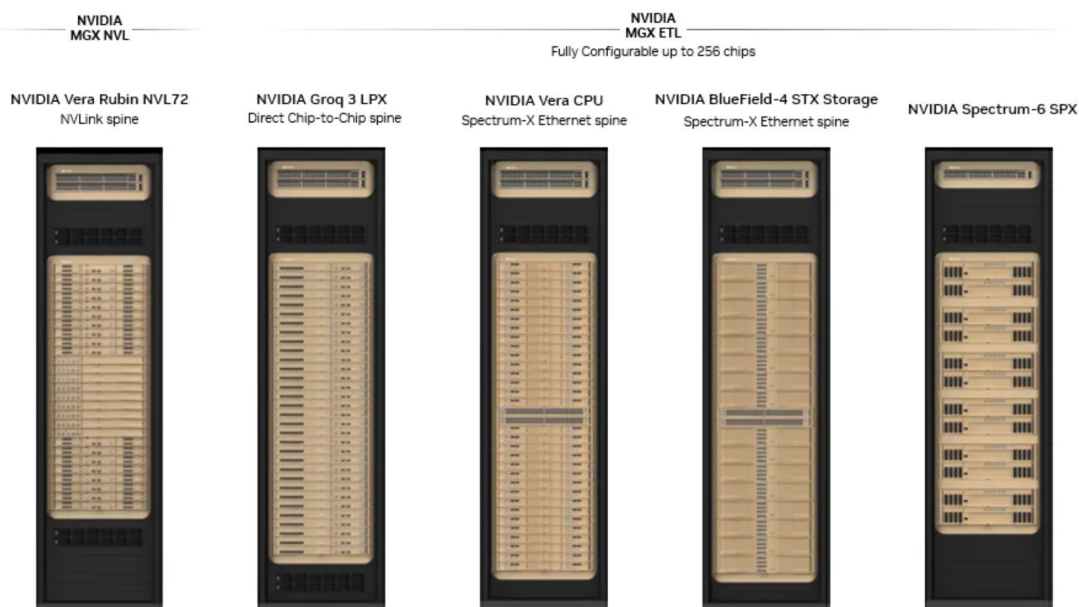
Vera Rubin now in full production, lowest token cost

NVDA announced Vera Rubin is now ramping into full production, across its five-rack system spanning Vera Rubin NVL72 compute, Vera CPU, Groq 3 LPX, Spectrum-6 SPX Ethernet, and Vera BlueField-4 STX storage racks.

Per NVDA, the full system delivers up to 10x higher inference perf per watt and 10x lower cost per token, and up to 35x higher throughput per watt for trillion-parameter models when paired with Groq 3 LPX. In a world of limited power, throughput per watt directly translates to revenue, and NVDA's extreme co-design and system-level solution can provide the lowest token cost by up to x factors. NVDA also noted the Vera Rubin platform is being scaled by hundreds of supply-chain ecosystem partners, with the supply chain 2x as large as Grace Blackwell's.

Exhibit 16: NVDA will begin offering standalone Vera CPU racks, alongside the Vera Rubin Platform beginning in 2H26, more directly addressing previously unaddressed or overlooked workloads in AI inference

Nvidia Vera Rubin SuperPOD with 5 different types of racks making up one giant 40-rack pods



Source: Nvidia

Vera CPU: head node vs. agentic AI likely 50/50 opp'ty

Traditional CPUs are increasingly becoming bottlenecks to agentic GPU utilization, directly affecting token throughput and latency. Vera CPU is designed to address this issue, with its 88x custom NVDA Olympus (Arm) cores all working together in tandem with the greatest cross-sectional bandwidth (3x bandwidth per core, 2x total bandwidth vs. x86), vs. traditional CPUs being rented core by core.

While Vera was initially designed as a head/compute node and falls short in raw core counts vs. competition (at 100-200+ cores per CPU), we flag NVDA's extreme codesign of full end-to-end system can help offset some of the disadvantages in even agentic workloads (where core/thread counts matter). Notably, the Vera CPU supports NVDA's **Spatial Multi-threading (SMT)**, vs. many other x86 products having offered multi-threading of some sort (INTC Hyperthreading, AMD simultaneous multithreading).

NVDA expects ~\$20bn of Vera CPU sales in F2H27, roughly split evenly between head/compute node sockets (traditional GPU support role in NVL racks) and standalone agentic AI sockets (reinforcement learning, orchestration roles in CPU racks).

NVDA has shipped 2.5mn Grace CPUs to date, and the follow-on Vera shipment could already reach ~4-5mn units in just the first two quarters of launch, at an estimated ASP of \$4-5k per CPU. NVDA also noted that the Vera builds on Grace's existing system-level software and optimization on NVDA tech stack, a key differentiation vs. first-gen Grace. Overall, the total CPU unit demand could potentially be higher than GPU over time, reinforcing the CPU-GPU ratio argument of 1-to-1 across both racks (vs. 1-to-2 and 1-to-4 prior).

NVDA content \$/GW rising materially gen-over-gen

NVDA believes every new generation of its systems can increase the company's addressable market materially, from the current ~\$40bn/GW for **Blackwell Ultra**, toward **\$60-80bn/GW for Vera Rubin and Rubin Ultra**, and toward ~\$100bn/GW for **Feynman** maybe. While exact figures may vary, NVDA continues to address more parts of the AI system each gen, with the current Vera Rubin solution comprised of 7 different chips across 5 different racks (vs. Blackwell just 1 main compute rack), and Rubin Ultra (2027) expected to add another rack on top. Beginning the Feynman generation (2028), NVDA also plans to offer optional optical scale-up (via CPO) for customers wanting a larger NVLink domain (vs. regular copper up to 576-die or 144 GPU packages).

"King of diversity": ACIE as important as hyperscale/cloud

NVDA's sales mix between ACIE (AI Clouds, Industrial, Enterprise) and hyperscale is ~50/50 today, with ACIE likely the faster growing segment over time. NVDA is the "king of diversity", as it supplies to every hyperscale, every major model maker, and has significant incumbency and moat in ACIE-type workloads (we view custom XPU's have limited opportunities here). NVDA's continuous upgrades to system software (i.e. Hopper/Blackwell) and new model rollouts are also key competitive advantages.

RTX Spark for personal agentic AI, for top 10% Windows

NVDA and Microsoft unveiled RTX Spark, a new superchip for Windows PCs purpose-built for personal AI agents and expanding NVDA's platform from cloud AI factories into personal AI endpoints. Built on TSMC 3nm, the chip offers 1 petaflop of AI perf., enabling 120bn-parameter LLMs with up to 1mn-token context locally. RTX Spark combines a Blackwell RTX GPU with 6,144 CUDA cores with a 20-core Grace CPU over NVLink-C2C, with MediaTek collaborating on the custom CPU design. Critically, we flag the chip compares to the Apple Mac Mini (10-12 CPU cores) or MacBook Pro (15-18 CPU cores), and likely targets the top 10% of the Windows market at ~25-30mn units/yr

We maintain Buy, \$350 PO, and a top AI pick, given attractive 16x CY27 PE or 0.4x PEG vs. Mag-7 avg of 1.9x.



QCOM: U/P on competitive landscape

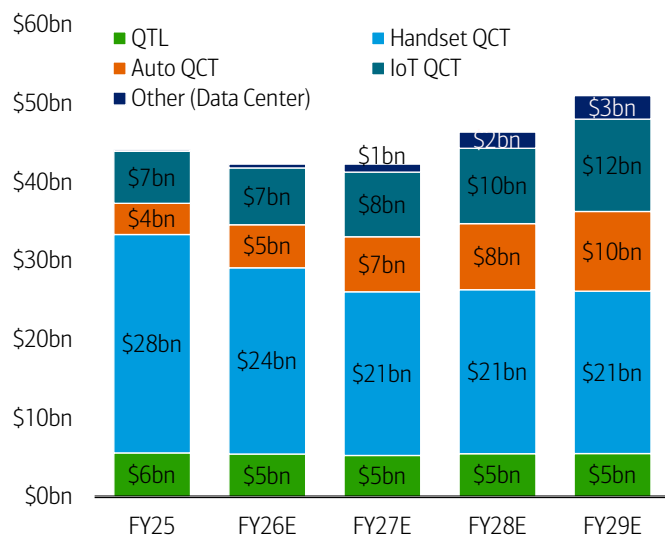
We reiterate Underperform on QCOM. In our view, the core handset franchise faces multi-year headwinds as rising memory prices further soften the weak handset market, while Apple modem in-sourcing removes high-margin revenue stream, and China OEM concentration heightens pricing and geopolitical risks.

Against that backdrop, the bull case is increasingly driven by QCOM's diversification into auto/IoT and data center business potentials. While we await further detail at upcoming Jun-24 AI-focused Investor Day, we see compute initiatives years away from being financially material, carry meaningful execution risk, and will require material investment to compete head-on with far better-resourced incumbents INTC, AMD, ARM, and NVDA.

While QCOM may only need to capture a small portion of the ~\$170bn+TAM to lift the stock, we see sustained share gain momentum as difficult amid intensifying competition. QCOM's ability to leverage their existing high-volume agreements with TSMC and Samsung for new data center products may work in their favor, however.

Exhibit 17: QCOM only expected to grow at a 4% CAGR between FY25-FY29E due to handset market softness and Apple roll-off

Summary of QCOM segment revenue forecast

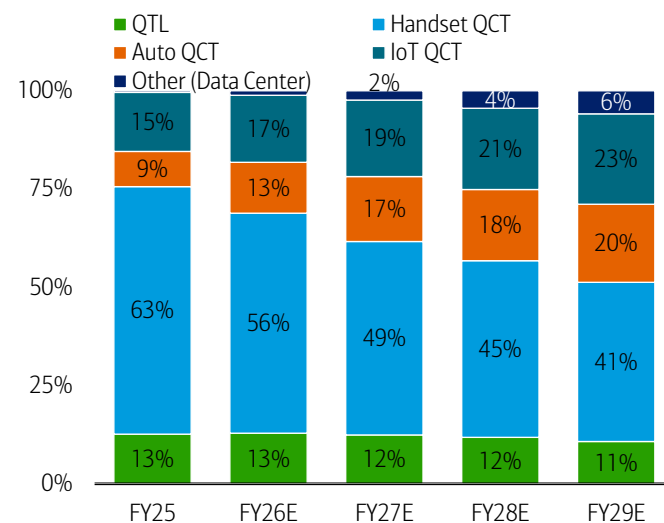


Source: BofA Global Research Estimates

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Exhibit 18: QCOM diversification is underway, though Data Center revenues likely only \$3bn or ~6% by FY29E

Summary of QCOM segment mix forecast



Source: BofA Global Research Estimates

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QCOM's Accelerator Effort: AI200 and AI250

QCOM's upcoming AI200 (2026) and AI250 (2027) rack-scale inference accelerators lean on Hexagon NPU IP and power efficiency expertise from mobile while using lower cost LPDDR memory. QCOM has a ~200MW commitment from HUMAIN beginning late 2026.

We remain constructive on QCOM's engineering capabilities but with inference now the most contested part of the accelerator TAM, a 2026 start provides QCOM with no ecosystem advantage vs. NVDA, AMD, AVGO/GOOGL, and MRVL/AWS while other hyperscalers are already developing high-cost ASICs for their respective clusters.

Data Center CPU re-entrance to leverage Arm-compatible Oryon core

QCOM's current CPU push rests on Arm-compatible Oryon core acquired via Nuvia (2021) which has only shipped at scale in Snapdragon X PC silicon. Porting a laptop-class core to a competitive server SKU poses significant execution challenges while most hyperscalers, neoclouds, and sovereigns already have established server CPU partners.



Role of CPUs in Agentic AI

For much of the AI training era, the compute narrative was dominated by GPUs and custom accelerators. That was appropriate: training large frontier models is highly parallel, matrix-heavy, and accelerator-bound.

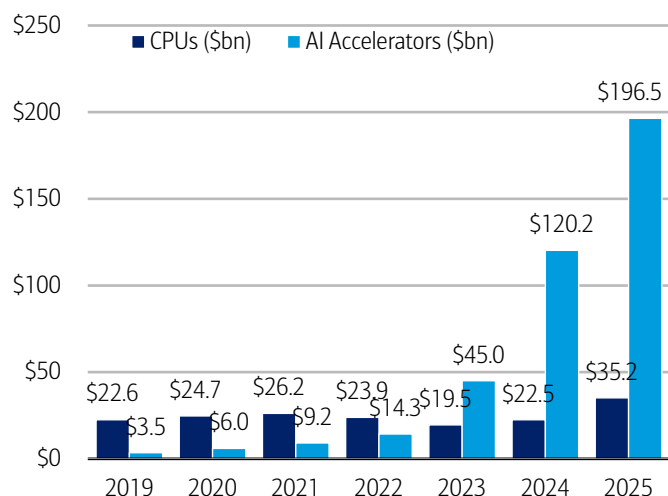
In that regime, **CPUs were necessary but largely secondary** — responsible for feeding accelerators, preprocessing data, tokenization, scheduling, storage access, and orchestration.

As a result, AI accelerator spend grew far faster than CPU spend particularly between 2022 and 2025, and accelerators had become the overwhelming majority of data center compute spend, while CPUs represented a much smaller share. The AI accelerator market had grown at +139% CAGR due a tremendous increase in training demand, vs. server CPUs only +14% CAGR.

By 2025, AI accelerators have become 85% of total compute spend and CPUs just 15%, versus accelerators at just 26% (CPUs 74%) in 2021 when the training market had just begun to emerge.

Exhibit 19: AI accelerator market grew at a +139% CAGR between 2022-2025 from a tremendous rise of AI training demand, vs. CPUs at +14% CAGR

CPUs vs. AI Accelerators TAM (\$bn)

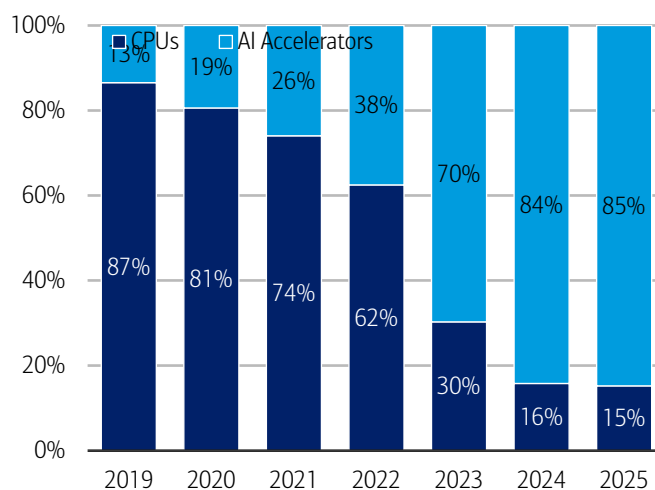


Source: BofA Global Research estimates, Mercury Research

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Exhibit 20: AI accelerators as a % of total data center compute spend has risen to 85% by 2025 (CPUs 15%), from just 26% in 2021 (CPUs 74%) before the rise of AI compute demand

CPUs vs. AI Accelerators TAM (%)



Source: BofA Global Research estimates, Mercury Research

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CPUs in training

Overall in training, CPUs are responsible for decompression, tokenization, batching, and ultimately feeding the GPUs. CPUs' main job is to make sure the GPUs/accelerators are utilized as much as possible by constantly preparing and queuing batches, making sure the expensive accelerators are fully in work without bottlenecks.

CPUs control the orchestration and preprocessing of data, i.e. they are typically not involved in the actual heavy processing that AI training requires (mostly done in parallel processing by GPUs/accelerators).

CPUs essentially perform a secondary or supporting role for the GPUs/accelerators in AI training, hence explaining the decline in the % of value in the overall system (vs. GPUs).



CPU in inference

In AI inference, the role of CPUs has become more broad and extensive.

The AI inference process can be broken into three phases: load phase, prefill phase, and decode phase.

1. In **load or initialization**, models are loaded from the disks into the memory, with performance largely dependent on overall disk (I/O) and CPU speed. Weight loading, quantization, and compilation all occur during the load phase, with the latter two (quantization and compilation) much better to be done offline vs. online to avoid runtime overhead.

Overall, the role/speed of CPU is key during the load phase of AI inference.

2. In **prefill**, time to first token (TTFT) and KV cache creation are the two main roles, with both prefill time and first token latency heavily dependent on overall compute/large matrix multiplication abilities, i.e. GPU/accelerators and networking rather than CPUs.

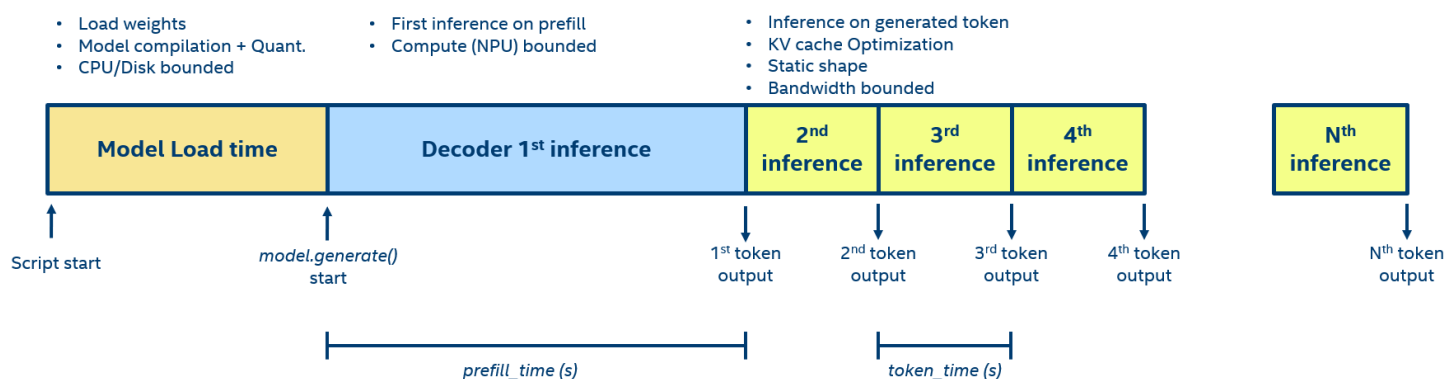
However, the CPUs do provide a role in tokenization, routing, batching, and memory setup.

3. In **decode**, performance is heavily impacted by KV cache reuse, so optimizing KV cache and/or increasing the bandwidth/capacity of system memory is critical, as such with high-bandwidth SRAM or HBM memory, inference storage context memory (i.e. low-latency storage), or even CXL memory (pooling of system memory) methodology.

Here too, increasingly the role of CPUs is increasing in decode because of scheduling, token-by-token control flow, sampling, guardrails, logits processing, memory management, and KV cache routing.

Exhibit 21: LLM inference process can also be broken into three phases: load phase, prefill phase, and decode phase

LLM Inference Process in Three Phases



Source: Intel

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CPUs in Agentic AI

In the next phase of AI – production inference, agentic AI, enterprise AI – is structurally even more CPU-intensive.

Agentic AI does not simply ask a model one question and receive one answer. It plans, retrieves context, calls tools, manages state, invokes APIs, interacts with databases, routes tasks across models, evaluates intermediate outputs, applies guardrails, and loops until a task is completed.

The practical implication is that CPUs become **more than “host processors”**. They become the **control plane** of AI inference.

In a conventional LLM inference request, the GPU may dominate the prefill and decode compute. But in agentic workflows, a single user request can fan out into many CPU-bound subtasks.

Intel’s agentic AI commentary describes these systems as multi-step workflows where CPUs manage orchestration, multiple agents, tool integration, state across workflows, and parallel decision trees. Ciena similarly argues that agentic AI shifts infrastructure beyond a “GPU-only” narrative because orchestration, memory handling, and tool execution increase the role of CPUs and networks in distributed AI workflows.

Exhibit 22: CPUs excel in document processing/classification, indexing, and captioning, among other AI workloads

Role of CPUs in AI Inference

AI Inference Workload	Good fit for...		
	CPUs	CPUs + PCIe-Based GPU	GPU Clusters
Document processing and classification	✓		
Data mining and analytics	✓		✓
Scientific simulations	✓		
Translation	✓		
Indexing	✓		
Content moderation	✓		
Predictive maintenance	✓		✓
Virtual assistants	✓	✓	
Chatbots	✓	✓	
Expert agents	✓	✓	
Video captioning	✓	✓	
Fraud detection		✓	✓
Decision-making		✓	✓
Dynamic pricing		✓	✓
Audio and video filtering		✓	✓
Financial trading			✓
Telecommunications and networking			✓
Autonomous systems			✓

Source: AMD

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Agentic AI is system-led, not CPU or GPU-led

However, this does not mean CPUs displace GPUs.

We view that agentic AI rather increases the importance of balanced compute:

- GPUs and accelerators remain essential for matrix math, large-model inference, high-throughput prefill, dense decoding, and frontier-model reasoning.

As we see in current-gen NVDA GB300 NVL72-style racks or next-gen VR200 NVL72-style racks/pods, accelerator-heavy architectures remain central today, with GPU throughput and memory capacity still acting as bottlenecks in many workloads.

In other words, we expect CPUs to rise in importance alongside accelerators, not instead of them, with **system bottlenecks broadening** to scheduling, memory management, data movement, retrieval, tool latency, networking, and concurrency.

AI CPUs expand system TAM, not replace

We see two distinct pieces to the AI CPU opportunity:

1. **Host CPUs** inside GPU/accelerator racks, which scale with accelerator deployments. This is the traditional approach and role of CPUs where they sit in 1 CPU per 2 GPU configuration in modern day compute accelerator racks.
2. **Incremental CPU-only agentic racks** for RAG, tool execution, small/mid-model inference, data processing, scheduling, vector DB/memory services, enterprise workflows, and agent orchestration.

Exhibit 23: We see increased significance of CPUs in agentic inference workloads, though we see the role of GPUs/accelerators to remain unchanged

Role of CPUs/GPUs in AI Training vs. AI Inference

Dimension	AI Training	Agentic AI Inference	Implications
CPU role	Data prep, tokenization, scheduling, feeding GPUs	Planning, routing, batching, tool execution, API calls, memory management, KV cache routing	Higher CPU attach and potential standalone CPU racks
GPU role	Training and high-throughput compute	Repeated model calls, prefill, decode, reasoning	GPU demand remains intact / expands with more inference calls
Dominant workload	Large parallel matrix compute	Multi-step reasoning, retrieval, tool use, state management	CPUs and networking matter more, but accelerators remain essential
Primary bottleneck	GPU FLOPS, HBM bandwidth, accelerator scale-up	GPU inference + CPU orchestration + memory / I/O / tool latency	Broader infrastructure TAM

Source: BofA Global Research

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We see the largest new opportunity for CPUs is in agentic orchestration and memory-rich inference infrastructure.

Even though a traditional GPU compute rack already contains CPUs (currently generally 1 CPU per 2 GPU configuration), agentic AI can create additional demand for CPU-dense infrastructure outside the GPU rack – its own **CPU-only racks**.

CPU-only racks

These CPU racks may host RAG pipelines, embedding services, vector databases, data preparation, API/tool execution, and small and mid-sized model inference, etc. Particularly, Lenovo and Intel describe CPU-only AI-agent and RAG inference as viable for a meaningful class of enterprise workloads, especially smaller and mid-sized language models, using Intel Xeon 6 and AMX acceleration.

Red Hat and Intel similarly argue that inference can use smaller, more cost-effective CPU platforms for certain functional and performance requirements, while CPUs remain the platform of choice for data processing, analytics, and classical ML.

Essentially, **CPU racks expand the TAM** by addressing niche/tail workloads that had previously been too small or inefficient for expensive GPU compute racks to address.

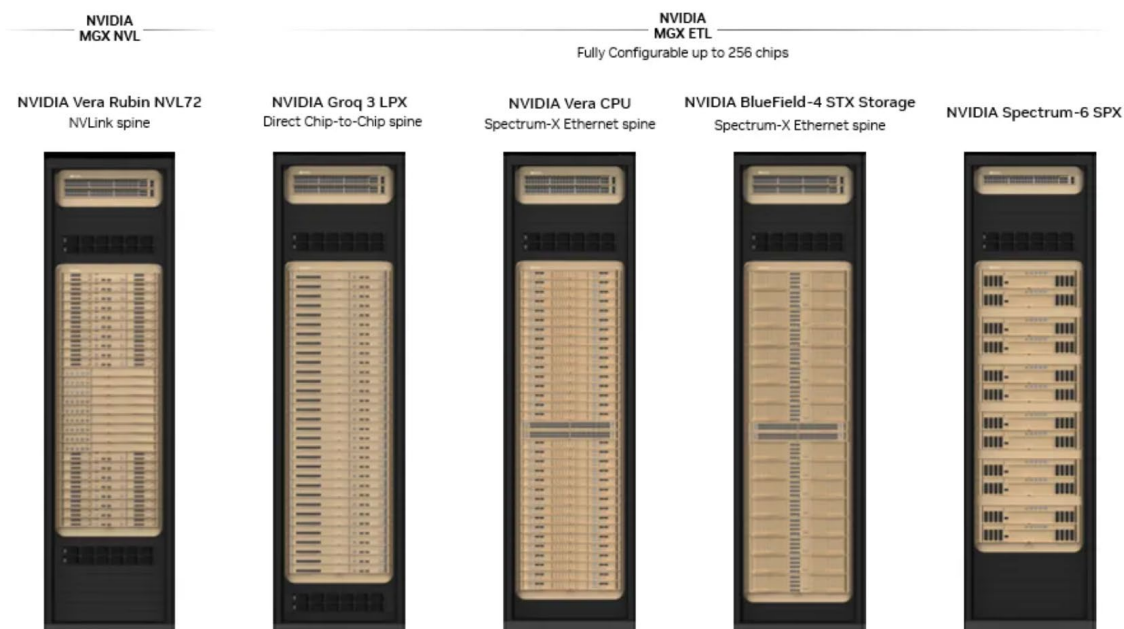


This is in-line with NVDA's announcement of its latest Vera CPU-only rack, to launch alongside the Vera Rubin Platform in 2H26.

The NVDA Vera CPU rack integrates up to 256 Vera CPUs in a dense, liquid-cooled rack to provide scalable, energy-efficient capacity. Per NVDA, a single rack can test, execute, and validate results from the Vera Rubin NVL72 (compute) and LPX (ultra-low-latency) racks. Ultimately, these Vera CPU racks provide the foundation for large-scale agentic AI and reinforcement learning.

Exhibit 24: NVDA will begin offering standalone Vera CPU racks, alongside the Vera Rubin Platform beginning in 2H26, more directly addressing previously unaddressed or overlooked workloads in AI inference

Nvidia Vera Rubin SuperPOD with standalone Vera CPU rack



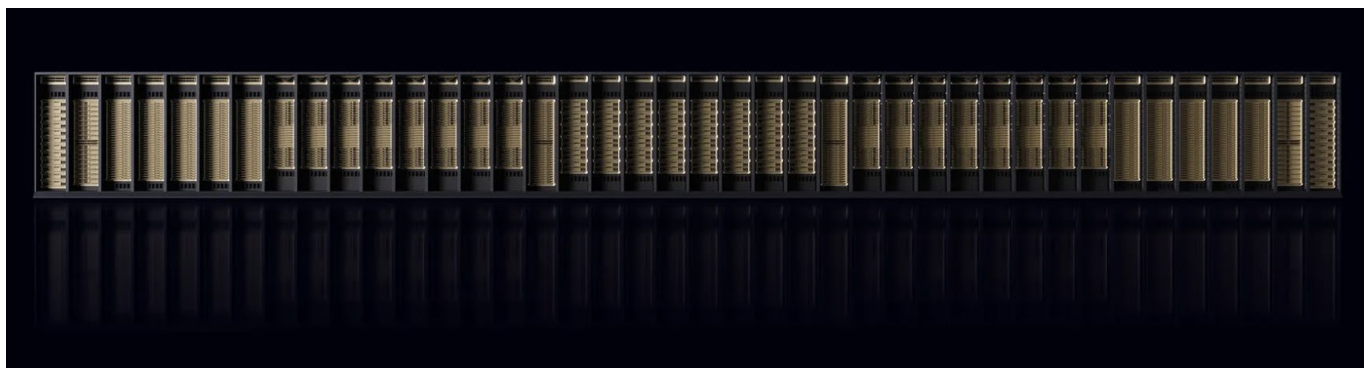
Source: Nvidia

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When stretched out to the entire 40-rack pod cluster, each Vera Rubin POD contains 5 different rack-scale systems (as seen above): 1) 1,152 Rubin GPUs (across 16 racks), 2) 576 Vera CPUs (as part of the GPU compute racks), and 3) potentially another 512 Vera CPUs (as part of standalone CPU racks, assuming 2 racks per pod). This would total 1,088 CPUs to 1,152 GPUs, reaching the roughly **1-to-1 ratio in agentic AI systems**.

Exhibit 25: Each Vera Rubin pod contains 1,152 Rubin GPUs, 512 Vera CPUs (as part of compute), and potentially another 576 Vera CPUs (standalone)

Nvidia Vera Rubin SuperPOD, full pod of 40 racks



Source: Nvidia

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Exhibit 26: Stocks mentioned in this report

Stocks mentioned

BofA Ticker	Bloomberg Ticker	Price	PO	Rating
AMD	AMD US EQUITY	US\$ 452.4	\$560.00	C-1-9
ARM	ARM US EQUITY	US\$ 307.43	\$335.00	C-2-9
INTC	INTC US EQUITY	US\$ 107.04	\$135.00	C-1-9
NVDA	NVDA US EQUITY	US\$ 200.42	\$350.00	C-1-7
QCOM	QCOM US EQUITY	US\$ 191.2	\$165.00	B-3-7

Source: BofA Global Research

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Glossary

- ACIE — AI Clouds, Industrial, Enterprise
- AI — Artificial Intelligence
- AI200 — Qualcomm AI200
- AI250 — Qualcomm AI250
- AMX — Advanced Matrix Extensions
- AMD — Advanced Micro Devices
- API — Application Programming Interface
- A/P — Advanced Packaging
- ARM — Arm / Arm Architecture
- ASIC — Application-Specific Integrated Circuit
- ASP — Average Selling Price
- AWS — Amazon Web Services
- BF16 — Brain Floating Point 16-bit
- CAGR — Compound Annual Growth Rate
- CAPEX — Capital Expenditure
- CCG — Client Computing Group
- CDNS — Cadence Design Systems
- COGS — Cost of Goods Sold
- CPU — Central Processing Unit
- CPO — Co-Packaged Optics
- CUDA — Compute Unified Device Architecture
- CXL — Compute Express Link
- CY — Calendar Year
- DB — Database
- DC — Data Center
- DCAI — Data Center and AI
- DIMM — Dual Inline Memory Module
- DRC — Design Rule Check
- DTCO — Design Technology Co-Optimization
- E — Estimate
- EBIT — Earnings Before Interest and Taxes
- EBT — Earnings Before Taxes
- EDA — Electronic Design Automation



- EMIB — Embedded Multi-die Interconnect Bridge
- EMIB-T — Embedded Multi-die Interconnect Bridge-T
- EM-IR — Electromigration / IR Drop
- EPS — Earnings Per Share
- ETL — Extract, Transform, Load
- EV/S — Enterprise Value to Sales
- FP16 — 16-bit Floating Point
- FY — Fiscal Year
- GDS — Graphic Data System
- GenAI — Generative Artificial Intelligence
- GM — Gross Margin
- GOOGL — Alphabet / Google
- GPU — Graphics Processing Unit
- GTC — GPU Technology Conference
- GW — Gigawatt
- HBM — High Bandwidth Memory
- HF — High Frequency
- HPC — High Performance Computing
- IaaS — Infrastructure as a Service
- IDC — International Data Corporation
- IDM — Integrated Device Manufacturer
- INTC — Intel
- INT8 — 8-bit Integer
- I/O — Input / Output
- IoT — Internet of Things
- IP — Intellectual Property
- KV cache — Key-Value Cache
- LLM — Large Language Model
- LPDDR — Low-Power Double Data Rate
- LPX — NVIDIA LPX Rack Label
- MGX — NVIDIA MGX Platform
- MRDIMM — Multiplexed Rank Dual Inline Memory Module
- MSFT — Microsoft
- MW — Megawatt
- N/A — Not Available / Not Applicable
- NCI — Non-Controlling Interest
- NCI Adj. — Non-Controlling Interest Adjustment
- NoSQL — Non-Structured Query Language
- Non-GAAP — Non-Generally Accepted Accounting Principles
- NPU — Neural Processing Unit
- NVDA — NVIDIA
- NVL — NVLink
- NVL72 — NVLink 72
- OS — Operating System



- OpM — Operating Margin
- PCIe — Peripheral Component Interconnect Express
- PDK — Process Design Kit
- PE — Price to Earnings
- P/E — Price to Earnings
- PEG — Price / Earnings to Growth
- pf-EPS — Pro Forma Earnings Per Share
- PO — Price Objective
- POD — NVIDIA POD / SuperPOD Cluster
- PPA — Performance, Power, Area
- QCOM — Qualcomm
- QCT — Qualcomm CDMA Technologies
- QTL — Qualcomm Technology Licensing
- RAG — Retrieval-Augmented Generation
- RAM — Random Access Memory
- RAS — Reliability, Availability, Serviceability
- RL — Reinforcement Learning
- RTX — NVIDIA RTX
- RTL — Register Transfer Level
- RTL-to-GDS — Register Transfer Level to Graphic Data System
- SAM — Serviceable Available Market
- SBC — Stock-Based Compensation
- SerDes — Serializer / Deserializer
- SLC — System-Level Cache
- Slurm — Simple Linux Utility for Resource Management
- SMT — Simultaneous Multi-threading / Spatial Multi-threading
- SOP — Sum of Parts
- SOTP — Sum of the Parts
- SPX — NVIDIA SPX Rack Label
- SQL — Structured Query Language
- SRAM — Static Random Access Memory
- STX — NVIDIA STX Rack Label
- TAM — Total Addressable Market
- TCO — Total Cost of Ownership
- TPU — Tensor Processing Unit
- TTFT — Time to First Token
- UALink — Universal Accelerator Link
- UCle — Universal Chiplet Interconnect Express
- U/P — Underperform
- XPU — Accelerator Processing Unit / Generic AI Accelerator
- YoY — Year over Year
- 2.5D — Two-and-a-Half-Dimensional Packaging
- 3D — Three-Dimensional Packaging
- 14A — Intel 14 Angstrom Process Node



- 14A-E — Intel 14 Angstrom Enhanced Process Node
- 18A — Intel 18 Angstrom Process Node
- 18A-P — Intel 18 Angstrom Performance Process Node

Price objective basis & risk

Advanced Micro Devices, Inc (AMD)

Our \$560 PO is based on 42x our 2027E non-GAAP EPS. Our PO basis is now towards middle/upper range of historical 13x-58x, but is well supported by AMD's 50%+ annual EPS CAGR potential and its AI CPU/GPU share gain potentials, modestly offset by slower growth in cyclical PC/embedded/console markets.

Downside risks: 1) Execution on first rack-scale product (MI400 Series), 2) Timing/Magnitude of Middle East AI Projects, 3) Lumpy nature of consumer and enterprise spending that could create delays in acceptance and success of new products, 4) High reliance on one outsourced manufacturing partner, 5) Maturity of current game console cycle.

Upside risks are greater share gain potential in the PC and server processor market against competitors.

Arm Holdings (ARM)

Our \$335 PO is based on conceptual sum-of-parts valuation that values the IP business at \$229 (2.0x CY30E PEG, discounted back 2 years) and the chiplet business at \$123 (35x CY30E PE, discounted back 2 years), with both multiples in-line with competitive peer range/average. Our PO implies 95x total company CY28E PE which is still within ARM's historical 35x-147x range, given longer-term data center content and silicon/chiplet (AGI CPU) opportunities, offset by high SoftBank dependence and opex ramp.

Upside risks: 1) better than expected smartphone/PC/server unit shipment, 2) faster adoption of higher-royalty rate v9 and CSS IP at customers, 3) faster share gains and ramp of AGI CPU chiplet lineup, 4) further proliferation of AI data centers and hyperscale-specific custom products, 5) Qualcomm/Nuvia content expansion post-settlement, 6) small floating volume.

Downside risks: 1) historically cyclical nature of semiconductor units, 2) high exposure to mature smartphone market, 3) competition against established x86 and other custom Arm-based CPU offerings in the data center, 4) emerging competition from RISC-V in low-end consumer markets, 5) rising geopolitical tensions and deterioration of Arm China relationship, 6) ongoing Qualcomm/Nuvia litigation, 7) small trading float.

Intel (INTC)

Our \$135 price objective is based on 25x CY30E EPS power of \$6+, discounted back two years, given the company's key server CPU and external foundry wafer/packaging opportunities that extend far out. Our PO implies 9.1x our total company 2028E EV/S which is well above historical 1.7x-4x range but we view as appropriate given increasing server CPU and external foundry opportunity, offset by near-/medium-term manufacturing ramp-up uncertainties.

Upside risks to our price objective are 1) key external foundry packaging/wafer deals that could significantly boost sales/utilization, 2) greater than expected yields/ramps at 18A and upcoming 14A nodes, resulting in greater GM/utilization profile, 3) stronger than expected PC market from Windows 10 refresh or AI uplift, 4) geopolitical tensions boosting sentiment for domestic manufacturing asset.



Downside risks to our price objective are 1) lower than yield/ramp at Intel Foundry, particularly for its new 18A and upcoming 14A nodes, 2) lack of material external foundry customer in wafer processing, 3) weaker-than-expected trends in a mature PC market, which is largest revenue generator for Intel, 4) accelerated share loss to major CPU competitors.

NVIDIA Corporation (NVDA)

Our \$350 PO is based on 26x CY27E PE ex cash, within NVDA's historical 25x-56x forward year PE range, which we believe is justified by NVDA's leading share in fast-growing AI compute/networking markets, offset by lumpiness in global AI projects, cyclical gaming market, and concerns around access to power.

Downside risks are: 1) weakness in consumer driven gaming market, 2) Competition with major public firms, internal cloud projects and other private companies in AI and accelerated computing markets, 3) Larger than expected impact from restrictions on compute shipments to China, or additional restrictions placed on activity in the region, 4) Lumpy and unpredictable sales in new enterprise, data center, and autos markets, 5) Potential for decelerating capital returns, and 6) Enhanced government scrutiny of NVDA's dominant market position in AI chips.

Qualcomm (QCOM)

Our PO of \$165 is based on 15x CY27E pf-EPS (ex-SBC). Our multiple is a discount to diversified/analog peers at 12x-25x and compute/AI peers at 17x-19x on QCOM's more muted growth outlook and relative lack of leverage to data center infrastructure buildout.

Downside risks: Top customer modem roll-off (\$7-8bn by Fall '27), further share loss at other key Android customer (100% to 75%), additional customer insourcing, QTL renewal risk (1H27), China exposure, DRAM tightness from AI server demand, competition, crowded AI data center CPU/accelerator market.

Upside risks: auto/IoT diversification towards \$22bn FY29E target, AI200/AI250 inference accelerators, Nuvia ARM CPUs, HUMAIN 200MW deployment, premium mix shift, rising on-device AI content.

Analyst Certification

I, Vivek Arya, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

US - Semiconductors and Semiconductor Capital Equipment Coverage Cluster

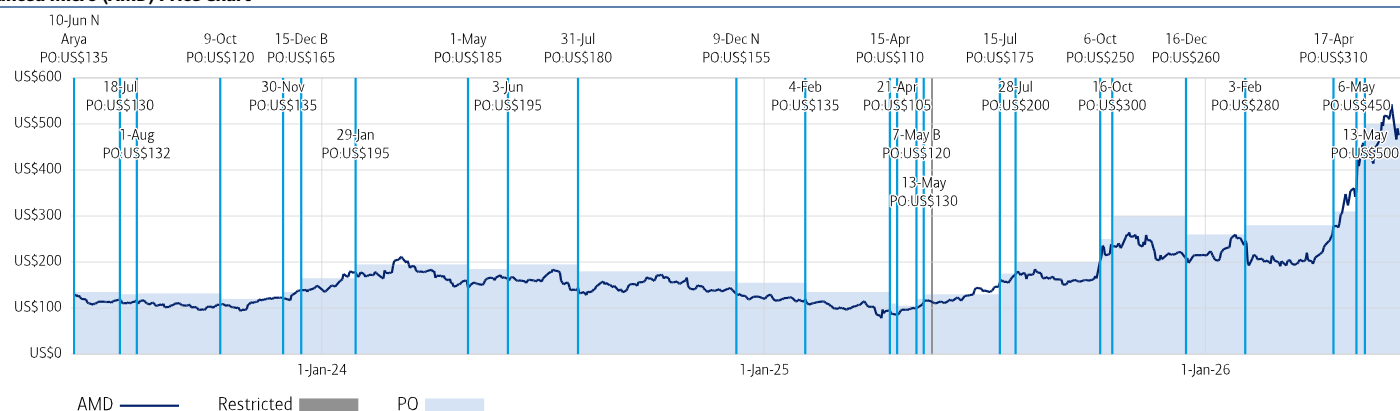
Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Advanced Energy Industries	AEIS	AEIS US	Duksan Jang
	Advanced Micro Devices, Inc	AMD	AMD US	Vivek Arya
	Allegro Microsystems	ALGM	ALGM US	Vivek Arya
	Analog Devices Inc.	ADI	ADI US	Vivek Arya
	Applied Materials, Inc.	AMAT	AMAT US	Vivek Arya
	Broadcom Inc	AVGO	AVGO US	Vivek Arya
	Cadence	CDNS	CDNS US	Vivek Arya
	Camtek	CAMT	CAMT US	Michael Mani
	Credo Technology	CRDO	CRDO US	Vivek Arya
	Intel	INTC	INTC US	Vivek Arya
	KLA Corporation	KLAC	KLAC US	Vivek Arya
	Lam Research Corp.	LRCX	LRCX US	Vivek Arya
	M/A-Com	MTSI	MTSI US	Vivek Arya
	Marvell Technology, Inc.	MRVL	MRVL US	Vivek Arya
	Microchip	MCHP	MCHP US	Vivek Arya
	Micron Technology, Inc	MU	MU US	Vivek Arya
	MKS Instruments	MKSI	MKSI US	Michael Mani
	Nova	NVMI	NVMI US	Michael Mani
	NVIDIA Corporation	NVDA	NVDA US	Vivek Arya
	onsemi	ON	ON US	Vivek Arya
	Synopsys	SNPS	SNPS US	Vivek Arya
	Teradyne	TER	TER US	Vivek Arya
	Texas Instruments Inc.	TXN	TXN US	Vivek Arya
NEUTRAL				
	Ambarella	AMBA	AMBA US	Vivek Arya
	Ambiq Micro, Inc.	AMBQ	AMBQ US	Vivek Arya
	Arm Holdings	ARM	ARM US	Vivek Arya
	Astera Labs Inc	ALAB	ALAB US	Vivek Arya
	Coherent Corp	COHR	COHR US	Vivek Arya
	Lumentum Holdings	LITE	LITE US	Vivek Arya
	NXP Semiconductors NV	NXPI	NXPI US	Vivek Arya
UNDERPERFORM				
	Axcelis Technologies	ACLS	ACLS US	Duksan Jang
	GlobalFoundries	GFS	GFS US	Vivek Arya
	Lattice Semiconductor	LSCC	LSCC US	Duksan Jang
	Qualcomm	QCOM	QCOM US	Vivek Arya
	Skyworks Solutions, Inc.	SWKS	SWKS US	Vivek Arya
RVW				
	Wolfspeed Inc	WOLF	WOLF US	Vivek Arya



Disclosures

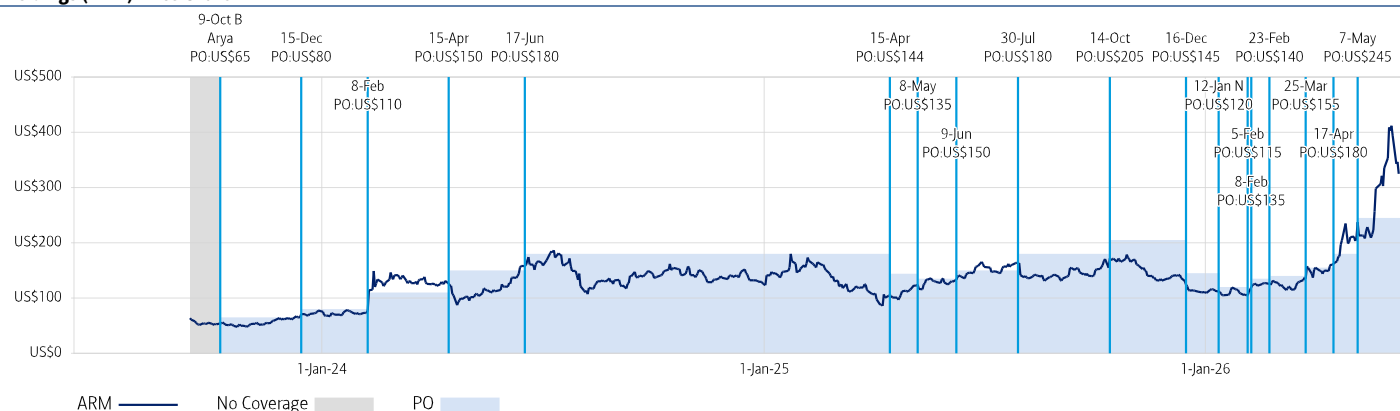
Important Disclosures

Advanced Micro (AMD) Price Chart



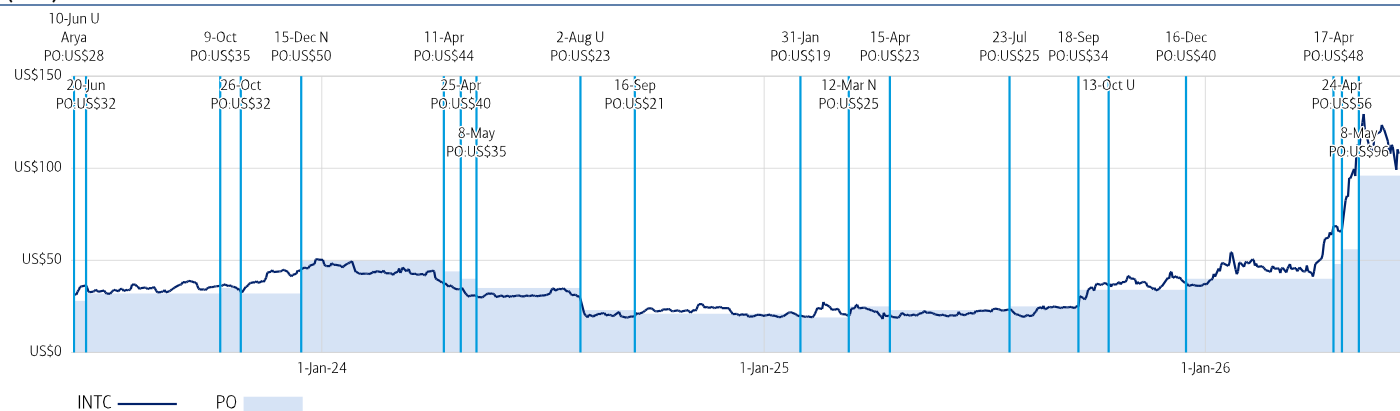
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Arm Holdings (ARM) Price Chart

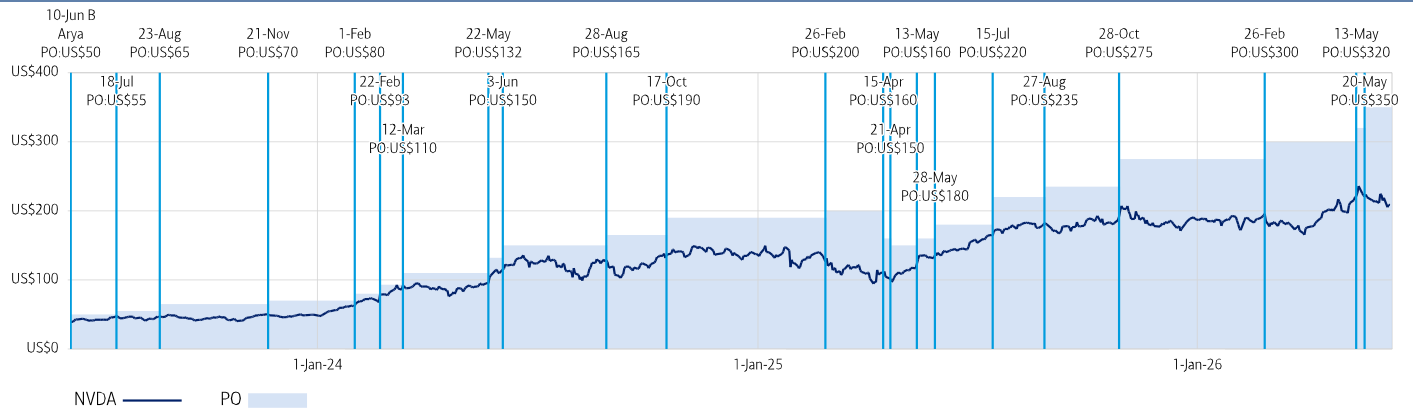


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Intel (INTC) Price Chart



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NVIDIA (NVDA) Price Chart

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Qualcomm (QCOM) Price Chart

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Equity Investment Rating Distribution: Technology Group (as of 31 Mar 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	234	58.50%	Buy	123	52.56%
Hold	90	22.50%	Hold	43	47.78%
Sell	76	19.00%	Sell	23	30.26%

Equity Investment Rating Distribution: Global Group (as of 31 Mar 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	1993	55.76%	Buy	1186	59.51%
Hold	821	22.97%	Hold	509	62.00%
Sell	760	21.26%	Sell	400	52.63%

^{R1} Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.



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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster ^{R2}
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Underperform	N/A	≥ 20%

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